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Dynamics of transcriptional programs and chromatin accessibility in mouse spermatogonial cells from early postnatal to adult life

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Abstract

In mammals, spermatogonial cells (SPGs) are undifferentiated male germ cells in testis that are quiescent until birth and then self-renew and differentiate to produce spermatogenic cells and functional sperm from early postnatal life throughout adulthood. The transcriptome of SPGs is highly dynamic and timely regulated during postnatal development. We examined if such dynamics involves changes in chromatin organization by profiling the transcriptome and chromatin accessibility of SPGs from early postnatal stages to adulthood in mice using deep RNA-seq, ATAC-seq and computational deconvolution analyses. By integrating transcriptomic and epigenomic features, we show that SPGs undergo massive chromatin remodeling during postnatal development that partially correlates with distinct gene expression profiles and transcription factors (TF) motif enrichment. We identify genomic regions with significantly different chromatin accessibility in adult SPGs that are marked by histone modifications associated with enhancers and promoters. Some of the regions with increased accessibility correspond to transposable element subtypes enriched in multiple TFs motifs and close to differentially expressed genes. Our results underscore the dynamics of chromatin organization in developing germ cells and complement existing datasets on SPGs by providing maps of the regulatory genome at high resolution from the same cell populations at early postnatal, late postnatal and adult stages collected from single individuals.

eLife assessment

This **useful** study reports datasets on gene expression and chromatin accessibility profiles of spermatogonia at different postnatal ages in mice. The supporting data are considered **incomplete**. This study may be of interest to biomedical researchers working on male germline stem cells and male fertility.

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Introduction

Spermatogonial cells (SPGs) are the initiators and supporting cellular foundation of spermatogenesis in testis in many species, including mammals. In the mammalian testis, the founding germ cells are primordial germ cells (PGCs), which give rise sequentially to different populations of SPGs: primary transitional (T1)-prospermatogonia (ProSG), secondary transitional (T2)-ProSG and then spermatogonial stem cells (SSCs) (McCarrey, 2013; Rabbani et al., 2022; Tan et al., 2020). The ProSG population is exhausted by postnatal day (PND) 5 (Drumond et al., 2011) and by PND6-8, distinct SPGs subtypes can be distinguished on the basis of specific marker proteins and regenerative capacity (Cheng et al., 2020; Ernst et al., 2019; Green et al., 2018; Hermann et al., 2018; Tan et al., 2020).

SSCs represent an undifferentiated population of SPGs that retain regenerative capacity and divide to either self-renew or generate progenitors that initiate spermatogenic differentiation, giving rise to differentiating SPGs (diff-SPGs). Diff-SPGs form chains of daughter cells that become primary and secondary spermatocytes around PND10 to 12. Spermatocytes then undergo meiosis and give rise to haploid spermatids that develop into spermatozoa. Spermatozoa are then released into the lumen of seminiferous tubules and continue to mature in the epididymis until becoming capable of fertilization by PND42-48 in mice (De Rooij, 2017; Kubota and Brinster, 2018; Oatley and Griswold, 2017).

Recent work showed that SPGs in early postnatal life have distinct transcriptional signatures (Green et al., 2018; Hammoud et al., 2014; Hermann et al., 2018; Law et al., 2019). During the first week of postnatal development, SPGs display unique features necessary for the rapid establishment and expansion of the cell population along the basement membrane. This includes high expression of genes involved in cell cycle regulation, stem cell proliferation, transcription and RNA processing (Grive et al., 2019). In comparison, genes expressed in adult SPGs are involved in the maintenance of a balance between proliferation and differentiation and help constitute a steady cell population that ensures sperm formation across life. Previous transcriptome analyses have revealed that adult SPGs prioritize pathways related to paracrine signaling and niche communication, as well as mitochondrial functions and oxidative phosphorylation (Grive et al., 2019; Hermann et al., 2018). In addition, changes to chromatin accessibility, histone tail modifications and DNA methylation have also been reported in SPGs during postnatal development (Maezawa et al., 2017) (Hammoud et al., 2014; Hammoud et al., 2015). Today however, the relationship between transcriptome dynamics and chromatin landscape in SPGs during the transition from early postnatal to adult stage has not been fully characterized.

We characterized the transcriptome and chromatin accessibility in SPGs during the transition from early postnatal to adult stages in mice. The results reveal extensive changes in transcription and chromatin accessibility in particular, increased accessibility at enhancer regions in adult compared to postnatal SPGs. Regions with changes in chromatin accessibility are enriched in binding motifs of several TFs that are differentially expressed between postnatal and adult stages, suggesting a possible role in changes in chromatin accessibility. Analyses of chromatin accessibility at transposable elements (TEs) identified previously uncharacterized changes at long terminal repeats (LTR) and LINE L1 subtypes between developing and adult SPGs. Together, these findings suggest a functional link between transcriptional dynamics and chromatin accessibility in SPGs during development and underscore the plasticity of genome organization in male germ cells.

Results

FACS enriches SPGs collected from postnatal and adult mouse testis

We collected testes from 8- and 15-day old pups (PND8 and 15) and from adult males and prepared cell suspensions from each animal by enzymatic digestion. The preparations were enriched for SPGs by fluorescence-activated cell sorting (FACS) using specific surface markers (Kubota et al., 2004) (**Fig. 1A, B**). The purity of sorted cells was evaluated by immunocytochemistry using PLZF (ZBTB16), a well-established marker for SPGs (Costoya et al., 2004). FACS enriched cell populations from 3-6% PLZF+ (before FACS) to 85-95% PLZF+ (**Fig. 1B**), suggesting high SPGs enrichment.

To characterize the molecular identity of enriched SPGs populations, we profiled their transcriptome at PND8, PND15 and adulthood by total RNA sequencing (RNA-seq) (n=6 for each group) and examined the expression of known SSCs and somatic markers (Leydig and Sertoli cells) (**Fig. 1C** and Fig. S1A, B). Classical SPGs markers such as *c-kit* (Schrans-Stassen et al., 1999), *Id4* (Helsel et al., 2017; Sun et al., 2015), *Lin28a* (Chakraborty et al., 2014; Wang et al., 2020), *Zbtb16* (Costoya et al., 2004; Song et al., 2020) and *Gfra1* (He et al., 2007) were robustly expressed in all SPGs samples at each developmental stage (**Fig. 1C** and Fig. S1A, B), indicating high purity of sorted cells. In contrast, low to negligible expression of somatic cells markers such as *Vim* (Bernardino et al., 2018) and *Tspan17* (Gewiss et al., 2021) for Sertoli cells, and *Fabp3* and *Hsd3b1* for Leydig cells (Sararols et al., 2021) (**Fig. 1C**) was detected. To further validate the samples purity, we manually curated a list of spermatogonial and somatic cell markers derived from recent single-cell RNA-seq datasets (Cao et al., 2021; Sararols et al., 2021) and determined their expression level in our datasets. Again, we detected robust and consistent expression of all reported spermatogonial markers and low expression of somatic cells markers in all samples at each developmental stage (Fig. S1A).

To evaluate the cellular composition of samples, we applied computational deconvolution to our bulk RNA-seq datasets, employing publicly available single-cell RNA-seq datasets (Cobos et al., 2023). Trained on high-quality RNA-seq datasets from pure or single-cell populations, deconvolution algorithms create expression matrices reflecting the cellular diversity in reference datasets. These cell type-specific expression matrices are subsequently used to determine the cellular composition of bulk RNA-seq samples. First, we assessed the cellular composition of all our RNA-seq libraries, using datasets generated by Hermann et al. (2018), which characterized the single-cell transcriptomes of various populations of SPGs and testicular somatic cells in early postnatal (PND6) and adult stages. This enabled us to analyze the composition of isolated SPGs but also to assess potential somatic cell contamination using a unified dataset source. Upon re-analyzing single-cell RNA-seq datasets, we identified distinct cell-type clusters, marked by specific cellular markers as reported in the original and subsequent studies (Fig. S2A,B). Then, CIBERSORTx generated gene-expression signature matrices and estimated the cell-type proportions within our 18 bulk RNA-seq libraries (Newman et al., 2019). Evaluation of our postnatal libraries (PND8 and 15) against a PND6 signature matrix revealed a predominant derivation from SPGs, with average estimated proportions of SSCs being 0.99 and 0.85 for PND8 and PND15 samples, respectively (Fig. S2F). Notably, the analysis of PND15 libraries also suggested the presence of additional SPGs types, including progenitors and differentiating SPGs (Fig. S2F), albeit at lower frequency. Similarly, evaluation of our adult RNA-seq libraries using an adult signature matrix, showed an average SSCs proportion of 0.82, indicating a primary derivation from SSCs (Fig. S2G). Consistent with the findings from PND15 libraries, our deconvolution analysis also suggests the presence of additional SPGs types, including progenitors and differentiating SPGs in adult samples (Fig. S2G). However, unlike in postnatal libraries, the

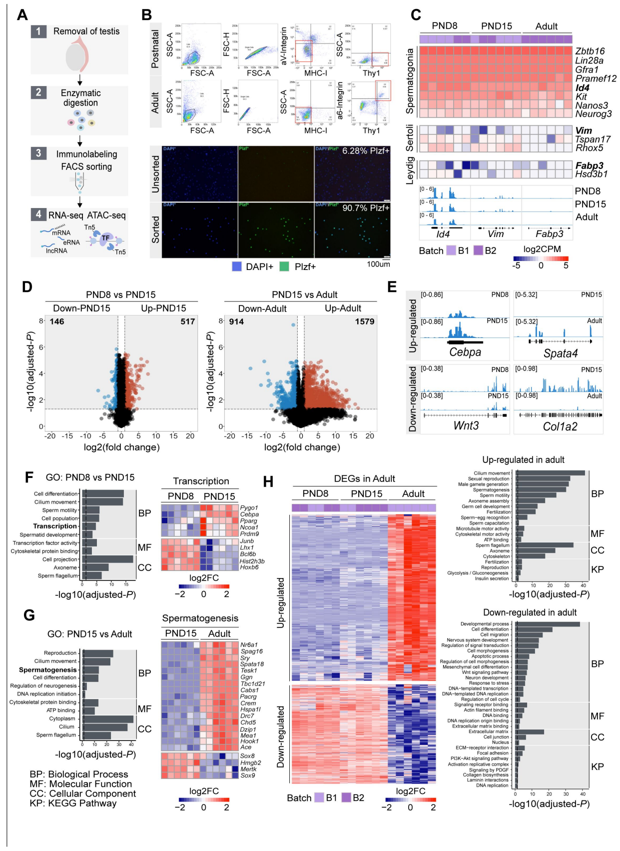

Figure 1.

Transcriptome dynamics between early postnatal and adult SPGs.

(A) Schematic representation of the experimental strategy to isolate and analyse SPGs from postnatal or adult male mice.

(B) Upper panel: Representative dot plots of the sorting strategy for enrichment of postnatal and adult SPGs. Gating based on side scatter/forward scatter (SSC-A/FSC-A) and forward scatter (height/forward scatter) (FSC-H/FSC-A) area was conducted to exclude cell debris and clumps. Lower panel: Representative immunocytochemistry on unsorted and sorted PND15 cells with anti-PLZF antibody and DAPI (nuclei) showing enrichment of PLZF+ cells after FACS.

(C) Heatmap of expression profile of selected markers of SPGs and different testicular somatic cells extracted from total RNA-seq data on PND8, PND15 and adult samples (n=6 for each group). Key genes for stem cell potential, stem and progenitor spermatogonia and Leydig and Sertoli cells were chosen to assess the enrichment of SPGs in the sorted cell populations. Each row in the heatmap represents a biological replicate from two experimental batches (B1 and B2). At the bottom are shown Integrative Genomics Viewer (IGV) tracks for aggregated RNA-seq signal for PND8, PND15 and adult over key genes used as markers. Gene expression is represented as Log2CPM (counts per million).

(D) Volcano plot of differentially expressed genes (DEGs) (adjusted-P≤0.05 and absolute Log2FC≥1) between PND8 and PND15 (left) and PND15 and adult (right). Figures in grey boxes indicate the number of down- and up-regulated DEGs in each comparison.

(E) Genomic IGV snapshots of exemplary DEGs showing aggregated RNA-seq signal for PND8, PND15 and adult SPGs.

(F) Left: Bar-plot of GO categories enriched in DEGs between PND8 and PND15 (adjusted-P≤0.05). Dotted line indicates threshold value for significance of 0.05. Right: Heatmap of DEGs between PND8 and PND15 belonging to the GO category "Transcription". Each row represents a biological replicate from two experimental batches. Shown are Log2FC with respect to average at PND8.

(G) Left: Bar-plot of GO categories enriched in DEGs between PND15 and adult (adjusted-P≤0.05). Dotted line indicates threshold value for significance of 0.05. Right: Heatmap of DEGs between PND8 and PND15 belonging to the GO category "Spermatogenesis". Each row represents a biological replicate from two experimental batches. Shown are Log2FC with respect to average at PND8.

(H) Left: Heatmap of all DEGs specific to adult SPGs. Each row represents a biological replicate from two experimental batches (B1 and B2). Shown are Log2FC with respect to average at PND8. Right, Bar-plots of GO categories enriched in up-regulated (adjusted-P≤0.05) (top) or down-regulated (adjusted-P≤0.05) (bottom) genes in adult SPGs. Dotted line indicates threshold value for significance of adjusted-P=0.05.

deconvolution analysis of adult libraries indicated the presence of other cell types (labelled "Other"), not corresponding to the major somatic cell types identified by Hermann et al. (2018). The estimated average proportion of these cells was less than 0.05 in two adult libraries and 0.10 in the others. This variance in cellular composition underlines the effectiveness of the deconvolution method to dissect the complex cellular composition of bulk RNA-seq samples. To further strengthen our observations, we re-analyzed two additional testicular single-cell RNA-seq datasets derived from an early postnatal (PND7) (Tan et al, 2020) and adult (Green et al., 2018) stage (Fig. S2C-E). Evaluation of our libraries confirmed a derivation from germ cells, in particular SSCs (Fig. S2G, F). Therefore, the results of the deconvolution analyses and our immunofluorescence data showing 85-95% PLZF+ cells in our cellular preparations confirm the efficiency of our FACS-based enrichment method and underscore that our bulk RNA-seq libraries are mainly composed of SPGs. The deconvolution analyses also suggest a predominant cellular composition of SSCs and to a lesser degree of differentiating SPGs, while also detecting a small proportion of somatic cells (<0.10) in our adult RNA-seq libraries, which is expected given the use of an immunolabeling-based enrichment method.

Dynamic transcriptomic states characterize postnatal and adult SPGs

We examined the transcriptome dynamics of SPGs from postnatal to adult stages by identifying differentially expressed genes (DEGs) in the RNA-seq datasets. We used a stringent cut-off (adjusted-P≤0.05, abs Log_2 fold change (FC)≥1) on 17,000 genes expressed during at least one developmental stage (See Methods). A total of 663 DEGs were identified between PND8 and PND15 (146 down-regulated and 517 up-regulated) and 2483 DEGs between PND15 and adult stage (914 down-regulated and 1579 up-regulated) (**Fig. 1D-E** and Table S1). Consistent with previous reports (Hammoud et al., 2015), we observed a dynamic regulation of germ cell factors, transcription factors (TF) involved in core pluripotency pathways and signaling molecules important for self-renewal (Fig. S1B). For instance, *Pou5f1 (Oct4)*, a TF necessary for pluripotency (Nichols et al., 1998), is significantly down-regulated in adult SPGs while the TFs *Klf4* and *Sox2*, also needed for pluripotency (An et al., 2019), are expressed similarly in postnatal and adult stages although at different level (i.e *Klf4* is expressed more than *Sox2*) (Fig. S1B).

We conducted Gene Ontology (GO) enrichment analyses to identify the biological processes which DEGs are involved in. GO analyses showed that DEGs between PND8 and PN15 are involved in cell differentiation, cilium movement, sperm motility and transcription and include for instance, TFs such as *Junb*, *Hoxb6*, *Cebpa* and *Pparg* (**Fig. 1F** and Table S2). Further, genes coding for histone proteins such as histone H2B *Hist2h3b* and epigenetic modifiers like the methyltransferase *Pygo1* and histone acetyltransferase *Ncoa1* are also differentially expressed between PND8 and 15. Interestingly, genes down- or up-regulated between postnatal stages are involved in different processes. While down-regulated genes are implicated in cell differentiation, cell migration and regulation of proliferation for instance, *Cdkn1a* that is down-regulated at PND15 regulates genetic diversity during spermatogenesis (Kanatsu-Shinohara et al., 2022), up-regulated genes are rather involved in germ cell development, cell signalling, insulin secretion and transcription (Fig. S1C and Table S2). Further, DEGs between PND15 and adulthood play roles in reproduction, spermatogenesis, cell differentiation, DNA replication, glycolysis and extracellular matrix (ECM)-receptor interaction pathways (**Fig. 1G**, Fig. S1D and Table S2). For example, the DNA methyltransferase *Dnmt1*, necessary for epi/genome regulation (Edwards et al., 2017) and *Col4a2*, a subunit of type IV collagen and component of the basal membrane (Reissig et al., 2019), are specifically down-regulated in adult SPGs (Fig. S1C). Interestingly, expression of collagen has been associated to a high proliferative potential and the ability to form germ cell colonies in SPGs (He et al., 2005), suggesting that regulation of collagen genes in adult SPGs may decrease germ-stem cell potential. Genes up-regulated in adult SPGs are involved in cilium movement, germ cell development and glycolysis (Table S2).

We then examined the differences and similarities of transcriptional profiles across the three developmental stages. To identify unique or common changes in gene expression during the transition from early postnatal to adult stages, we compared the list of DEGs at PND8 versus PND15 and at PND15 versus adulthood. Remarkably, the vast majority of DEGs show significant stage-specific changes in transcription (Fig. S2A). For instance, 75% (495/663) of DEGs between PND8 and PND15 are not statistically changed in the transition to adult SPGs (Fig. S2B and Table S1 and S2). Similarly, 93% (2325/2493) of DEGs in adult SPGs are not statistically changed when compared to PND8 versus PND15 SPGs (**Fig. 1H** and Table S1 and S2). GO enrichment analyses showed that adult-specific down-regulated genes are involved in cell differentiation, cell migration, regulation of signal transduction and Wnt signalling, regulation of DNA replication and transcription as well as collagen biosynthesis and laminin interactions while adult-specific up-regulated genes are involved in sexual reproduction, male gamete generation, germ cell development, cilium movement and glycolysis (**Fig. 1H** and Table S1 and S2).

Interestingly, a small fraction of all DEGs (4.9%) are detected as significantly changed in the same direction and magnitude (adjusted-P≤0.05, $absLog_2FC≥1$) across the three developmental stages (Fig. S2C and Table S1 and S2). For instance, 121 genes are consistently up-regulated from PND8 to adult stage, while just 26 are consistently down-regulated from PND8 to adult stage (Fig. S2C). Such DEGs are involved in different biological pathways like chromatin organization (Table S2). In particular, the histone gene clusters displayed significant down-regulation across postnatal development and in adulthood (Table S2).

Landscape of chromatin accessibility in SPGs during postnatal development

Cellular differentiation is generally accompanied by changes in chromatin accessibility at regulatory elements (Atlasi et al., 2017). We examined how chromatin accessibility in SPGs is modified during development using omni-ATAC-seq (Corces et al., 2017). We focused on SPGs at PND15 and adulthood, stages that showed the highest changes in gene expression (**Fig. 1**). Accessible regions were identified by peak-calling on merged nucleosome-free fragments (NFF) as proxy for genomic regions with potential regulatory activity. We identified 158,977 peaks with clear ATAC-seq signal compared with surrounding genomic regions (Fig. S3A and Table S3). Most accessible regions were located at distal intergenic regions (38%), introns (28%) and putative promoter regions (23%) (Fig. S3B, C) and encompassed sequences enriched for histone PTMs associated with active chromatin such as H3K4 mono-, di- and tri-methylation (Fig. S2D). Notably, signal enrichment was higher for H3K4 methylation than for other histone PTMs (Fig. S3D). These results are consistent with previous observations that ATAC-seq peaks are identified at regulatory elements such as enhancers and promoters and are preferentially located at genomic regions with nucleosomes carrying H3K4me (Henikoff et al., 2020).

We then examined differences in chromatin accessibility between PND15 and adult SPGs by differential accessibility analysis. 3212 differentially accessible regions (DARs) were identified (adjusted-P≤0.05, $absLog_2FC≥1$) with a total of 760 regions with decreased chromatin accessibility (DARs-down) and 2452 regions with increased accessibility (DARs-up) in adult SPGs (**Figure 2A**, B and Table S3). DARs were predominantly localized in distal intergenic regions (54% DARs-down, 37% DARs-up) and introns (33% DARs-down, 36% DARs-up) with a minority located in putative promoter regions (8% DARs-down, 12% DARs-up), consistent with the genomic distribution of all detected ATAC-seq peaks (**Fig. 2C**). GO analyses of the closest genes assigned to each DAR showed that these genes are involved in male gonad development, cell adhesion, sex differentiation and regulation of cell communication among others (**Fig. 2D, E**).

Epigenomic annotation of DARs using high-quality publicly available ChIP-seq datasets for postnatal SPGs (Cheng et al., 2020) revealed that DARs are predominantly located in regions enriched for histone PTMs associated with enhancers and promoters. 51% of regions with

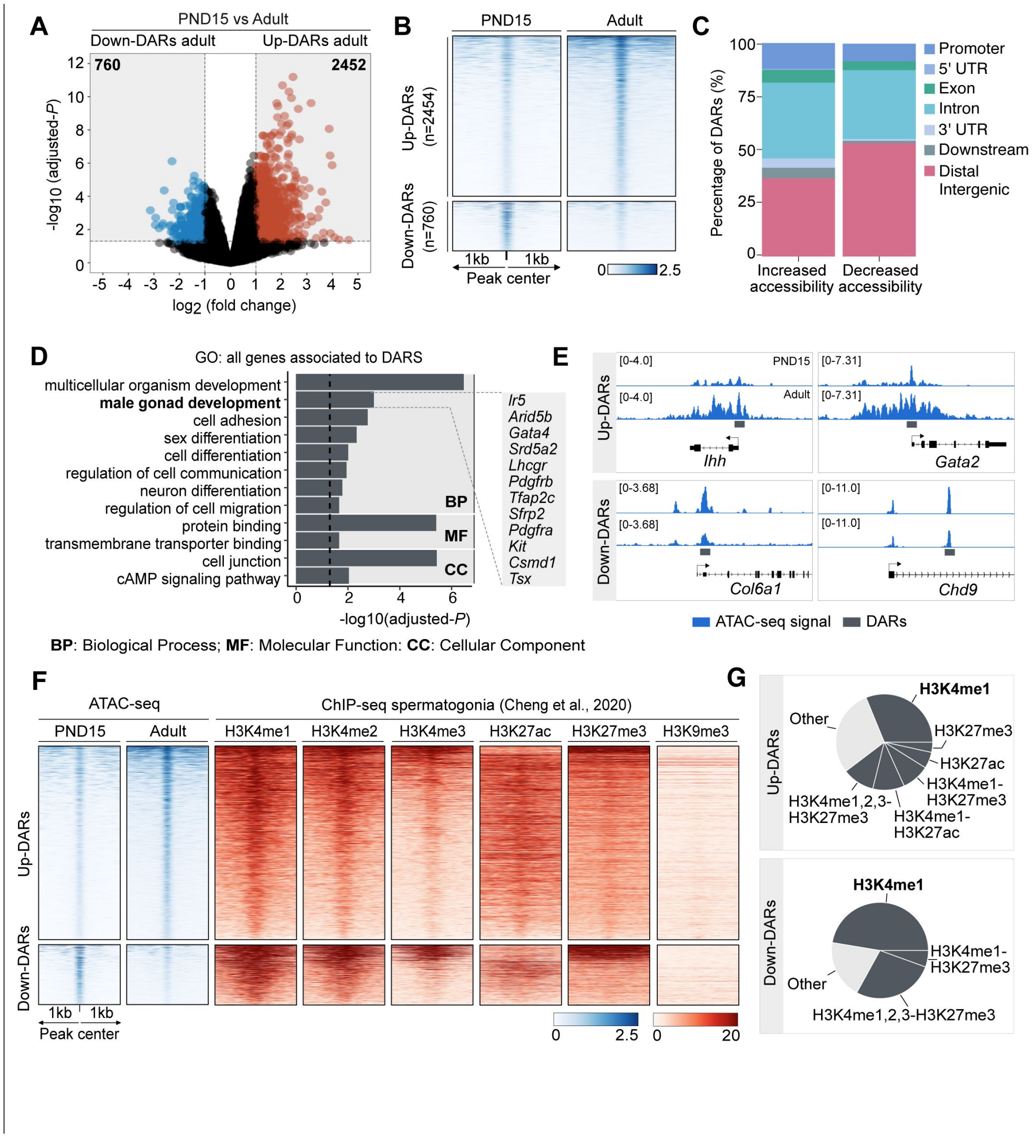

Figure 2.

Dynamics of chromatin accessibility of SPGs during the transition from postnatal to adult stage (A) Volcano plot of DARs obtained by ATAC-seq (adjusted-P≤0.05 and absolute Log2FC≥1) between PND15 and adult SPGs. Figures in grey boxes indicate the number of DARs with decreased (Down) or increased (Up) chromatin accessibility in PND15 compared to adult SPGs.

(B) Heatmaps showing normalized ATAC-seq signal for all identified DARs comparing PND15 and adult stages. Each row represents a 2kb genomic region extended 1kb down- and upstream from the centre of each identified DAR (as shown in A). Rows are ordered in a decreasing level of mean accessibility.

(C) Bar plot illustrating the genomic distribution of DARs between PND15 and adult SPGs.

(D) Bar plot of GO categories for associated genes assigned by proximity to all DARs. Dotted line indicates threshold value for significance of adjusted-P=0.05.

(E) IGV tracks for ATAC-seq signal for PND15 and adult SPGs showing DARs (red box) located at promoters and intergenic regions.

(F) Heatmaps showing normalized ChIP-seq signal for different histone marks in adult SPGs (public data derived from Cheng et al., 2020) at all identified ATAC-seq peaks from PND15 and adult stages. Each row in the heatmap represents a 2kb genomic region extended 1kb down- and up-stream from the centre of each identified peak. Shown data correspond to non-regenerative SPGs as stated in Cheng et al., 2020.

(G) Pie charts showing the overlap of all identified DARs with genomic regions significantly enriched for different histone marks in adult SPGs (public data derived from Cheng et al., 2020).

increased accessibility are located at genomic loci that carry histone PTMs since early postnatal stages, and about half (48%) overlap with regions significantly enriched in H3K4me1, a histone PTM associated to enhancers. In contrast, 80% of regions with decreased accessibility are located in regions with H3K4me1 and 1/3 (33%) also overlap with the repressive histone PTM, H3K27me3 (**Fig. 2F**). Interestingly, while 16% of regions with increased accessibility are located at potential active regulatory elements that carry H3K27ac, none of the regions with decreased accessibility overlap with H3K27ac (**Fig. 2F**). Therefore, our data indicate that the transition from postnatal to adult SPGs is accompanied by discrete, yet robust, changes in chromatin accessibility at potential regulatory elements, suggesting their involvement in the control of gene transcription.

DARs are associated with binding sites for distinct families of TFs

We examined the relationship between differences in chromatin accessibility and TF binding by TF motif analysis using DARs as input. We observed that regions with decreased accessibility in adult SPGs are enriched for binding motifs of members of specific TF families such as KLF (KLF2, KLF5, KLF11, KLF12, KLF15, KLF16), SP (SP1-5, SP9), FOXO (FOXO1, FOXO3), ETS (ETV1-6) among others (**Fig. 3A** and Table S4). Expression analyses showed that many of these TFs are differentially expressed in postnatal or adult SPGs (adjusted-P<0.05, 10 down-regulated and 11 up-regulated) (**Fig. 3B, C**). For example, KLF11,12 and 15, TFs that regulate stem cell maintenance and development (Bialkowska et al., 2017), are upregulated in adult SPGs compared to PND15 (**Fig. 3C**). These TFs and their motif match top enriched motifs detected in DARs in adult SPGs (**Fig. 3A**). The SP family of TFs also show differential expression and enrichment of binding motifs in regions with decreased accessibility. SP5, which has its lowest expression in adult SPGs (**Fig. 3B, C**), promotes self-renewal of mESCs by directly regulating *Nanog* (Tang et al., 2017) .

Members of the FOXO family of TFs also have an enrichment of motifs in regions with decreased accessibility. While FOXO3 has increased expression from PND15 onwards, FOXO1 is decreased in adult compared to postnatal SPGs (**Fig. 3B, C**). Interestingly, FOXO1 and FOXO3 regulate SSC function and maintenance in mouse (Goertz et al., 2011). Finally, ETS-type of TFs also show differential expression during SPGs development and their motif is enriched in regions with decreased accessibility. ELK4 is up-regulated in adult SPGs and can act as a transcriptional repressor via recruitment of the HDAC Sirt7 and deacetylation of H3K18ac (**Fig. 3B-D**) (Barber et al., 2012). EKL4 can also act as transcriptional activator of immediate early genes such as *c-Fos* (Dalton and Treisman, 1992). Interestingly, *c-Fos* itself codes for a TF important for the regulation of proliferation (He et al., 2008) and shows a trend towards up-regulation in adult SPGs (Table S1). In contrast, GABPA is downregulated in adult SPGs compared to PND15 and has been involved in the regulation of proliferation in mESCs (Ueda et al., 2017).

Regions with decreased chromatin accessibility also show an enrichment for the binding motif of the TF DMRT1. *Dmrt1* is progressively repressed during SPGs postnatal development and is the lowest in adult SPGs (**Fig 3B**, C and E). DMRT1 can act either as a repressor or activator and controls testis development and male germ cell proliferation (Zhang et al., 2016). DMRT1 can inhibit meiosis and promote mitosis in SPGs by repressing Stra8 (Matson et al., 2010). Consistently, we observed transcriptional repression of *Stra8* in adult SPGs (Table S1).

Next, we identified motifs overrepresented in regions with increased chromatin accessibility in adult SPGs. The identified motifs correspond to binding motifs of members of specific TF families such as POU (POU2F2, POU1F1, POU5F1), RFX (RXF1-7), DMRT (DMRT1, DMRT3, DMRTA1-2, DMRTC2), SOX (SOX2, SOX3, SOX5, SOX13), NFY (NFYA, NFYB, NFYC) and AP-1 (JUN, JUNB, JUND, FOS, ATF3, BATF) (**Fig. 3F** and Table S4). A subset of them is also differentially expressed (adjusted-P<0.05, 13 down-regulated and 5 up-regulated). The most overrepresented motif is similar to the binding motif of the POU family of TFs, which are critical regulators of stem cells (Nichols et al., 1998). *Pou1f1* and *Pou5f1* are transcriptionally repressed in adult SPGs while *Pou2f2* is maximally expressed in PND15 SPGs and down-regulated in adult cells (**Fig. 3G, H**). We also identified motifs for members of the RFX family of TFs, which are master regulators of

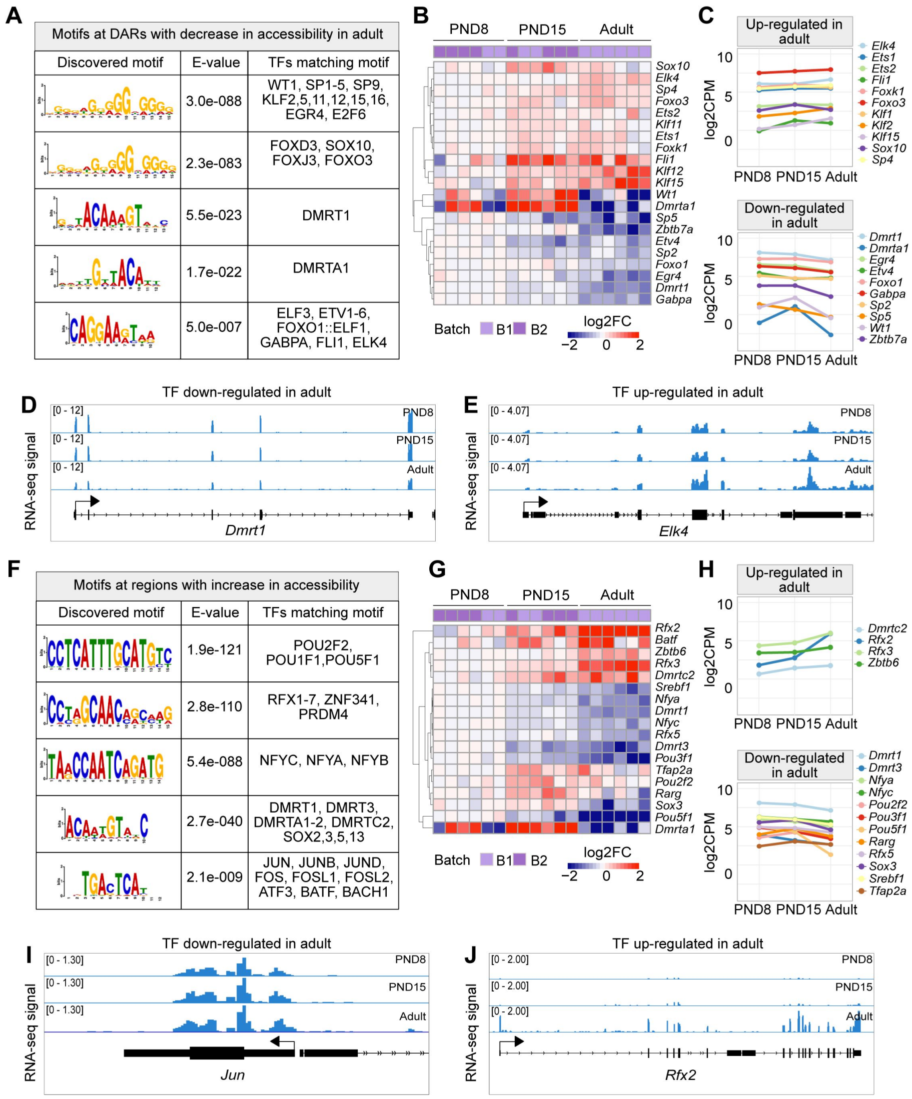

Figure 3.

TF binding motifs at DARs and their transcriptional dynamics during the transition from postnatal to adult SPGs

(A) Table of top five enriched motifs in genomic regions with decrease in chromatin accessibility in adult SPGs and TF matching motifs. A full list of TF motifs is provided in Table S4.

(B) Heatmap of expression profile of TFs with motifs in genomic regions with decreased chromatin accessibility shown in (A). Each row represents a biological replicate from two experimental batches. Unsupervised hierarchical clustering was applied to each row and a dendrogram indicating similarity of expression profiles among genes is shown. Shown are Log2FC with respect to average of PND8 SPGs.

(C) Line-plots of average expression of each gene displayed in the heatmap in (B) showing the dynamics of gene expression across PND8, PND15 and adult SPGs.

(D, E) Genomic snapshot from IGV showing aggregated RNA-seq signal from PND8, PND15 and adult SPGs for the TFs D) *Dmrt1* and E) *Elk4*.

(F) Table of top five enriched motifs in genomic regions with increase in chromatin accessibility in adult SPGs and TF matching motifs. A full list of TF motifs is provided in Table S4.

(G) Heatmap of expression profile of TFs with motifs in genomic regions with increase in chromatin accessibility shown in (F). Shown are Log2FC with respect to the average of PND8 SPGs.

(H) Line-plots of average expression of each gene displayed in the heatmap in (G) showing the dynamics of gene expression across SPGs development.

(I, J) Genomic snapshot from IGV showing the aggregated RNA-seq signal from PND8, PND15 and adult SPGs for the TFs I) *Jun* and J) *Rfx2*.

ciliogenesis (Choksi et al., 2014) implicated in regulation of neural stem cells (Kawase et al., 2014). RFX2 is robustly expressed in adult SPGs (**Fig. 3G, H** and **I**) and has been reported to induce the expression of ciliary genes in association with the TF FOXJ, which has a trend towards up-regulation in adult SPGs (Fig. S4) (Quigley and Kintner, 2017). Interestingly, ciliary genes are among the top genes specifically up-regulated in adult SPGs (**Fig. 1H**), suggesting a regulatory relationship between RFX2 and ciliary genes expression in adult SPG cells.

Other binding motifs enriched at regions with increased accessibility correspond to members of the NF-Y complex, NF-YA, NF-YB and NF-YC. In mESCs, NF-Y TFs facilitate a permissive chromatin conformation and play an important role in the expression of core ESC pluripotency genes (Oldfield et al., 2014). NF-YA/B motif enrichment has also been found in regions of open chromatin in human SPGs (Guo et al., 2017). Interestingly, the expression of NF-YA and NF-YC is progressively down-regulated starting at PND15 (**Fig. 3G, H**). We also detected an enrichment for the binding motifs of members of the AP-1 family of TFs, which are involved in many processes, from regulation of cell proliferation to differentiation and acute responses to environmental clues (Bejjani et al., 2019; Vierbuchen et al., 2017). Interestingly, except for *Fosl2*, other TFs do not show any significant change in transcription in postnatal or adult SPGs and are either constitutively expressed or repressed. For instance, c-FOS and JUN are robustly expressed in postnatal and adult SPGs (**Fig. 3J**) consistent with their role in promoting the proliferative potential of SSCs (He et al., 2008; Wang et al., 2018). Together, these data reveal that a shift in TFs repertoire accompanies chromatin reorganization occurring in SPGs during postnatal to adult development.

A subset of DARs is associated with differential gene expression

Thus far, our results show that in SPGs, the transition from postnatal to adult stage is accompanied by changes in gene expression and chromatin accessibility. In some instances, differences in gene expression can be correlated with differences in chromatin accessibility of regulatory elements. Since DARs in postnatal and adult SPGs overlap with putative enhancer and promoter elements, we examined if these changes in chromatin accessibility correlate with changes in gene expression. First, we tested if promoter regions of DEGs (+/-2.5kb from TSS) have differential chromatin accessibility by identifying all possible ATAC-seq peaks located in promoters and calculating an average accessibility signal. No significant difference was detected for down-regulated genes in adult SPGs, despite a trend for decreased accessibility around the TSS (**Fig. 4A**). However, down-regulated genes had a significant increase in chromatin accessibility in their body (**Fig. 4B**). Up-regulated genes had a significant increase in chromatin accessibility both at the promoter region and gene body (**Fig. 4A, B**). Then, we conducted motif enrichment analyses and identified families of TFs with overrepresented binding motifs at the promoter of DEGs. Binding motifs for members of KLF and SP family of TFs were overrepresented in the promoter of both down- and up-regulated DEGs, while motifs for members of TF families SOX and NFY were detected only in the promoter of down-regulated genes. Promoters of up-regulated genes were specifically enriched in motifs for RFX and AP-1 (**Fig. 4C**). These observations suggest that changes in chromatin accessibility, in particular at the promoter of up-regulated DEGs may be associated with differential binding of RFX and AP-1 TFs.

Next, we examined the overlap between DARs and the promoter of DEGs. Unexpectedly, only 3.2% of all promoters of DEGs overlap with at least one DAR (**Fig. 4D**). The majority of overlapping DARs (71%) is associated with promoters of up-regulated genes in adult SPGs (**Fig. 4D**). For example, *Gpx4*, a peroxidase involved in the control of stemness (Hu et al., 2021), is transcriptionally active in adult SPGs and has an open chromatin at its promoter region (**Fig. 4E**). Since the majority of DARs are located at distal intergenic regions and have histone PTMs associated with active regulatory elements, we extended our search of their target genes by assigning a large region (+/-100kb) around each DEG. We observed that a third of DEGs were associated with at least one DAR (**Fig. 4F**), with the majority of DARs (69%) overlapping regulatory elements of up-regulated genes. For instance, *Braf* is transcriptionally up-regulated in

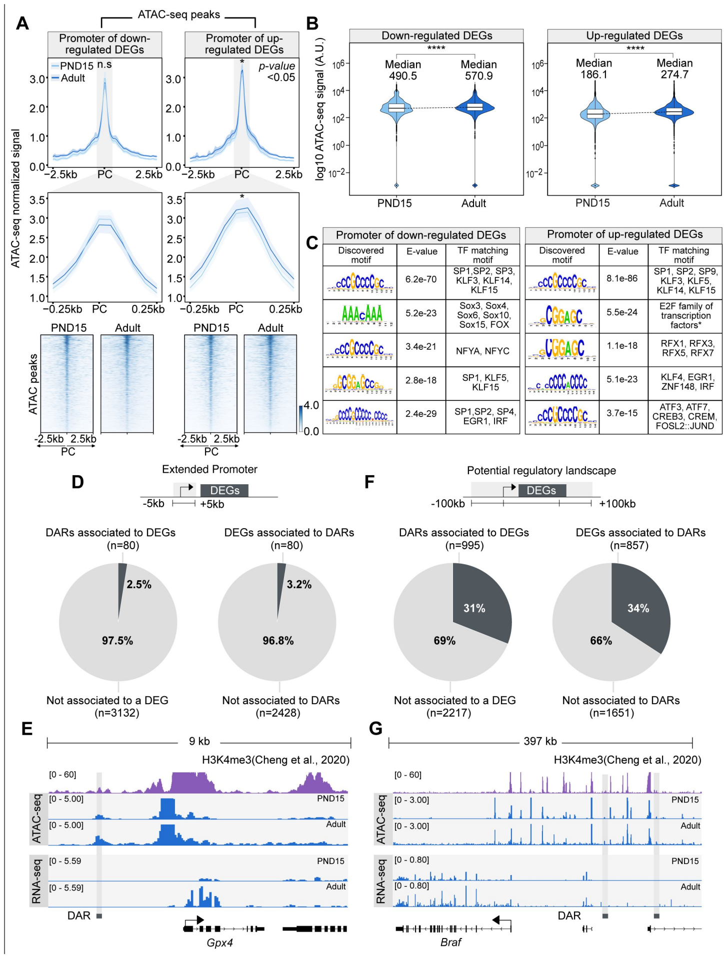

Figure 4.

Chromatin accessibility dynamics at DEGs during the transition from postnatal to adult SPGs

(A) *Top*, profile plots of ATAC-seq signal from PND15 and adult SPGs over all ATAC-seq peaks located within a genomic region of 5kb surrounding the TSS of DEGs. Mean (line) and bootstrap confidence interval (area, shadow lines) computed over all non-overlapping 50bp genomic regions. n.s, non-significant differences at TSS region. *, significant difference in signal between PND15 and adult at p-value<0.05. PC, peak center from ATAC-seq data. Profile plots for all ATAC-seq peaks surrounding the TSS of down-regulated (top left) or up-regulated (top right) DEGs. *Middle,* profile plots of ATAC-seq signal from PND15 and adult SPGs over all ATAC-seq peaks located within a genomic region of 500bp surrounding the TSS of DEGs. *Bottom*, heatmaps showing normalized ATAC-seq signal from PND15 and adult for all ATAC-seq peaks surrounding the TSS of all down-regulated (left) or up-regulated (right) DEGs. Each row in the heatmap represents a 2kb genomic region extended 1kb down- and up-stream from the center of each identified peak. Row Are ordered in a decreasing level of mean accessibility taking PND15 as reference.

(B) Violin plots of average ATAC-seq signal for PND15 and adult SPGs over gene body of down-regulated (left) or up-regulated (right) DEGs. ****p<0.0001.

(C) TF motif enrichment analysis for all ATAC-seq peaks surrounding the TSS of all down-regulated (left) or up-regulated (right) DEGs.

(D) *Left*, pie chart of percentage of DARs overlapping the extended promoter of DEGs (+/- 5kb from TSS). *Right*, pie chart of percentage of DEGs associated with a DAR.

(E) IGV tracks for an example of DAR associated to the extended promoter (+/- 5kb from TSS) of a DEG. Shown are tracks for RNA-seq and ATAC-seq signal for PND15 and adult SPGs generated in this study and signal track for H3K4me3 from SPGs derived from public ChIP-seq data (Cheng et al., 2020).

(F) *Left*, pie chart of percentage of DARs overlapping potential regulatory sequences of DEGs (+/-100kb from start and end of gene). *Right*, pie chart of the percentage of DEGs associated with a DAR.

(G) IGV tracks for an example of DAR associated with potential regulatory sequences (+/-100kb from start and end of gene) of a DEG.

adult SPGs, which correlates with increased chromatin accessibility at distal regulatory elements (**Fig. 4G**). Our results overall show that only a fraction of DEGs is associated with DARs, suggesting that changes in gene expression are not always mirrored at the level of chromatin accessibility, consistent with recent observations (Kiani et al., 2022).

Chromatin accessibility at transposable elements (TEs) undergoes remodeling during the transition from early postnatal to adulthood in SPGs

TEs are tightly regulated in the germline by coordinated epigenetic mechanisms involving DNA methylation, chromatin silencing and PIWI proteins-piRNA pathway (Deniz et al., 2019). SPGs were recently shown to have a unique landscape of chromatin accessibility at long-terminal repeats (LTRs) retrotransposons, compared with other stages of germ cells in testis (Sakashita et al., 2020). We examined chromatin accessibility at TEs in PND15 and adult SPGs by quantifying ATAC-seq reads overlapping TEs defined by UCSC RepeatMasker then analyzing differential accessibility at subtypes level (see Methods section). These analyses showed that SPGs transitioning from PND15 to adult stages have significantly different chromatin accessibility at 135 TEs subtypes (adjusted-P≤0.05, abs$\mathrm{Log_2}$FC≥1) (**Fig. 5A, B** and Table S5). Most TEs subtypes have decreased chromatin accessibility (93/135, 68,9%) (**Fig. 5A**) and 42 have increased accessibility (**Fig. 5B**) in adult SPGs. Differentially accessible TEs loci are predominantly located in intergenic (68%) and intronic (25%) regions and 6% re close to a gene (+/-1kb from TSS) (**Fig. 5C**).

LTR retrotransposons were the most abundant TEs with altered chromatin accessibility, specifically ERVK and ERV1 subtypes (**Fig. 5A, B**). Exemplary ERVK subtypes with less accessible chromatin included RLTR17, RLTR9A3, RLTR12B and RMER17B (Table S5). Enrichment of RLTR17 and RLTR9 repeats was previously reported in mESCs, specifically at TFs important for pluripotency maintenance such as *Oct4* and *Nanog* (Fort et al., 2014). Interestingly, we identified the promoter region of the long non-coding RNA (lncRNA) *Lncenc1*, an important regulator of pluripotency in mESCs (Fort et al., 2014; Sun et al., 2018), to harbor several LTR loci with decreased accessibility in adult SPGs. One of these loci, RLTR17, overlaps with the TSS of *Lncenc1*, and its decreased accessibility correlates with markedly lower *Lncenc1* expression in adult SPGs (**Fig. 5D**). *Lncenc1* (also known as *Platr18*) is part of the pluripotency-associated transcript (*Platr*) family of lncRNAs identified as potential regulators of pluripotency-associated genes *Oct4*, *Nanog* and *Zfp42* in mESCs (Bergmann et al., 2015). We identified several other *Platr* genes such as *Platr27* and *Platr14*, whose TSS overlaps with LTRs with reduced accessibility, RLTR17 and RLTR16B_MM, respectively (**Fig. 5D** and Table S5). These two pluripotency-associated transcripts tended to be downregulated in adult SPGs but were unchanged at PND8 and PND15 (**Fig. 5D** and Table S1). The other LTR subtypes with decreased accessibility in adult SPGs belong to ERV1, ERVL and MaLR families (**Fig. 5A**). A few non-LTR TEs also had decreased chromatin accessibility, particularly 7 DNA element subtypes, 2 satellite subtypes and 1 LINE subtype, respectively (Table S5).

TE subtypes with increased accessibility mostly included members of ERVK, ERVL and ERV1 families (24/42, 57.1%) (Table S5). Interestingly, many LINE L1 subtypes had increased chromatin accessibility in adult SPGs (Table S5). When parsing the data for more accessible loci within L1 subtypes, we found several L1 loci situated less than +/-5kb from the TSS of olfactory (*Olfr*) genes, particularly *Olfr* gene clusters on chromosome 2, 7 and 11 (Table S5).

TEs are known to provide tissue-specific substrates for TF binding (Sundaram and Wysocka, 2020). We examined the regulatory potential of differentially accessible LTR subtypes by determining the enrichment of TF motifs in these regions. We grouped LTR subtypes by family (EVK, ERV1, ERVL and ERVL-MaLR families) and observed that among families with less accessible chromatin, ERVKs had the highest number of enriched TF motifs in adult SPGs (Table S5). Top hits included TFs with known regulatory functions in cell proliferation and differentiation such as

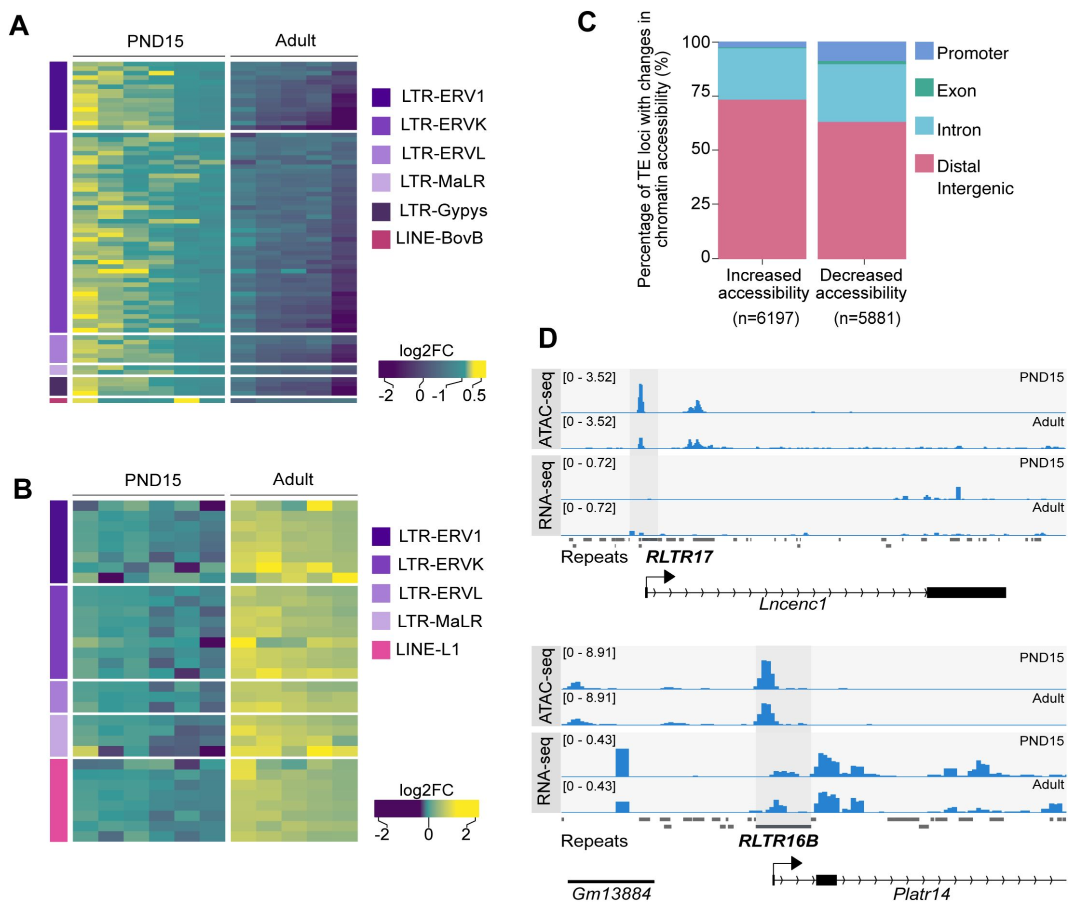

Figure 5.

Differential chromatin accessibility at TEs in PND15 and adult SPGs.
(A) Heatmap of LTR and LINE subtypes with decreased accessibility between PND15 and adult SPGs extracted from Omni-ATAC data (adjustedP≤0.05 and Log2FC≥0.5)
(B) Heatmap of LTR and LINE subtypes with increased accessibility between PND15 and adult SPGs extracted from Omni-ATAC data (adjustedP≤0.05 and Log2FC≥0.5). Expression changes of these subtypes between adult and PND15 SPGs RNA-seq from literature are represented as Log2FC.
(C) Bar plot illustrating the genomic distribution of DARs between PND15 and adult SPGs.
(D) Genomic snapshot from IVG showing aggregated ATAC-seq and RNA-seq signal from PND8, PND15 and adult SPGs for *Lncenc1* and *Platr14.* Below signal tracks is shown the RepeatMasker track for the mouse genome, and TEs with changes in chromatin accessibility are highlighted.

FOXL1 and FOXQ1, stem cell maintenance factors ELF1, EBF1 and THAP11 and TFs important for spermatogenesis PBX3, ZNF143 and NFYA/B (Table S5). ERVLs displayed motif enrichment for few TFs, among which ETV2, recently reported to be the SSC factor ZBTB7A and the testis-specific CTCF paralog CTCFL (Table S5) (Green et al., 2018).

More accessible LINE L1 subtypes were highly enriched in TF motifs, particularly in multiple members of ETS, E2F and FOX families (Table S5). The most significant motifs belonged to SSCs maintenance and stem cell potential regulators FOXO1 and ZEB1, as well as TFs recently associated with active enhancers of the stem cell-enriched population of SPGs such as ZBTB17 and KLF5 (Table S5) (Cheng et al., 2020). More accessible ERV1s were also enriched in several TF binding sites, including spermatogenesis-related TFs (PBX3, PRDM1, NFYA/B), hypoxia inducible HIF1A and cytokine regulators STAT5A/B, suggesting different metabolic demand between postnatal and adult stage (Table S5).

Discussion

This study examines the transcriptomic landscape and chromatin accessibility of mouse SPGs from early postnatal development to adulthood. Using FACS-enriched SPGs populations at PND8, PND15 and adulthood, we show that the transcriptome of SPGs is dynamically regulated during development and that many factors including pluripotency-related TFs and signaling molecules are differentially expressed. This correlates with changes in chromatin accessibility although not all DEGs have altered accessibility. TEs such as LTR retrotransposons have modified chromatin accessibility, which in some cases affects pluripotency-associated lncRNAs. TF motif enrichment in differentially accessible TEs suggests various regulatory roles for SPGs. Our findings complement existing datasets on spermatogonial cells by providing parallel transcriptomic and chromatin accessibility maps at high resolution from the same cell populations at early postnatal, late postnatal and adult stages collected from single individuals (for adults).

Different transcriptional states characterize the postnatal development of SPGs

SPGs are known to undergo genome-wide transcriptional changes during development (Grive et al., 2019; Hammoud et al., 2015; Hermann et al., 2018). The present data complement previous findings by showing that postnatal maturation of SPGs is accompanied by an overall decrease in the expression of pluripotency markers and differential expression of signaling molecules and regulators of cell cycle and spermatogenesis. The high quality and depth of our RNA-seq datasets allowed us to uncover specific transitional states of SPGs postnatal maturation that were not known before. For instance, we observed that while the transition from early to mid-postnatal stage (PND8 to PND15) is accompanied by changes in gene expression, the transition from mid-postnatal to adult stage is more extensive, both in number of regulated genes and magnitude of changes. We also observed a clear trend towards up-regulation of gene expression as postnatal maturation proceeds, that may reflect gradual transcriptional increase for some genes and possibly sharp changes at certain developmental stages for others.

Consistently, many transcriptional regulators and chromatin modifiers are differentially regulated during SPGs development. Different families of TFs including factors with a specific role in SPGs like RFX and DMRT, and factors with more widespread functions across cell types such as members of the AP-1 family are modulated, suggesting the contribution of multiple classes of TFs to SPGs development. Notably, several transcriptional repressors and epigenetic silencers are differentially down-regulated which may explain the marked trend towards transcriptional up-regulation in our datasets (**Fig. 1D**). Interestingly, we also observed a global down-regulation of histone genes starting at PND15 (Table S1), which may modify the composition and structure of chromatin and be associated with chromatin openness (Prado et al., 2017). Overall, our transcriptional data highlight the yet unappreciated dynamic regulation of TFs and epigenetic/chromatin modifiers in SPGs during postnatal development.

Chromatin accessibility in SPGs is modified at enhancers during postnatal development

Our results suggest that the transition of SPGs from early postnatal to adult stage is accompanied by changes in chromatin accessibility. In particular, accessibility is increased in regions enriched in H3K4me1, a histone PTM typically associated with primed enhancers, but also to a lesser extent, in regions enriched in H3K27ac, a mark associated with active regulatory elements (Gasperini et al., 2020; Shlyueva et al., 2014). This suggests that the transition of SPGs from postnatal to adult stage involves changes in the regulatory genome, in particular at primed enhancers. Enhancer priming has recently been proposed as an important mechanism for SPGs differentiation (McCarrey, 2023). Differences in chromatin accessibility may be partially explained by differential binding of TFs to regulatory elements (Klemm et al., 2019). This may involve at least three non-mutually exclusive events: 1) a change in the abundance of TFs, 2) a change in the epigenetic status of TF binding sites due to different amount and/or activity of epigenetic modifiers, and 3) a change in nucleosome positioning around TF binding sites due to different amount and/or activity of chromatin remodelers. For instance, CpG methylation and nucleosome positioning can influence TF binding at regulatory elements and affect chromatin accessibility (Barisic et al., 2019; Kaluscha et al., 2022). The fact that *Dnmt1* is downregulated in adult SPGs concomitantly with differential expression of TFs and chromatin modifiers may explain changes in chromatin accessibility at potential enhancers.

Modest correspondence between transcription and chromatin accessibility in SPGs

Despite robust changes in gene expression in adult SPGs, only a subset of DARs could be assigned to DEGs. This suggests that transcriptional regulation and chromatin accessibility at regulatory elements are not always linked and can be dissociated, as previously reported (Chereji et al., 2019; Kiani et al., 2022). This implies that chromatin accessibility may be modified without any effects on transcription and conversely, transcription can be activated or repressed without requiring any change in chromatin accessibility. In turn, it is possible that a repressor can be exchanged with an activator at a regulatory element without detectable consequences. It is also possible that changes in chromatin accessibility have no immediate effects on transcription but reflect an intermediate state that primes a gene or genomic locus for later activation or repression such as observed with developmentally- or experience-dependent primed enhancers (Marco et al., 2020; Rada-Iglesias et al., 2011). Consistently, the majority of DARs with increased accessibility overlap with regions enriched in H3K4me1, a histone modification characteristic of primed enhancers thought to control future transcriptional responses in different cell types (Rada-Iglesias et al., 2011). These regions may be important for responses to external cues in adult SPGs. The apparent lack of correspondence between transcription and accessibility may also be in part technical and due to the stringency of filters we used to assign genomic loci to DARs. We cannot exclude the possibility that modest changes in accessibility (<log2FC1) correspond to changes in TF occupancy. Chromatin accessibility has been suggested to not necessarily be the primary determinant of chromatin-mediated gene regulation (Chereji et al., 2019). Such possibility could be addressed by examining genome-wide maps of TFs, cofactors and RNA-Pol II occupancy that would provide more accurate information about the regulatory genome and its relationship with observed gene expression programs.

Chromatin accessibility at TEs during SPGs maturation

Our results reveal differences in chromatin accessibility and TFs motif landscape at different TEs subtypes between PND15 and adult SPGs. ERVK and ERV1 subtypes were the most abundant categories of TEs that become less accessible in adult SPGs, whilst LINE L1 subtypes gained accessibility. Although the majority of these TEs reside in intergenic and intronic regions, we could detect specific loci belonging to differentially accessible ERVK and LINE L1 subtypes localized nearby TSS of distinct gene families. Notably, ERVKs are known to play an important role in the regulation of mRNA and lncRNAs transcription during spermatogenesis (Davis et al., 2017; Sakashita et al., 2020). The landscape of chromatin accessibility at LTRs loci in SPGs is known to be unique compared to other mitotic germ cells and meiotic gametes in testis (Sakashita et al., 2020).

RLTR17, one of the LTR subtypes with decreased chromatin accessibility in adult SPGs overlaps with the TSS of several downregulated lncRNAs from the *Platr* family. *Platr* genes, including *Lncenc1* and *Platr14* identified in our study, are LTR-associated lncRNAs important for gene expression programs in ESCs (Bergmann et al., 2015). Interestingly, RLTR17 has also previously been linked to pluripotency maintenance. In mESCs, it is highly expressed and enriched in open chromatin regions and has been shown to provide binding sites for the pluripotency factors Oct4 and Nanog (Fort et al., 2014). Therefore, we suggest that RLTR17 chromatin organization may play a role in regulating pluripotency programs between early postnatal and adult SPGs.

In contrast to decreased accessibility at LTRs loci, LINE L1 subtype had increased accessibility in adult SPGs. Some of these L1 loci were located close to olfactory receptor genes with upregulated mRNA expression. Recent findings in mouse and human ESCs suggest a non-random genomic localization of L1 elements, specifically at genes encoding proteins with specialized functions (Lu et al., 2020). Among them, the *Olfr* gene family was the most enriched in L1 elements (Lu et al., 2020). Although their role in SPGs is currently not established, Olfr proteins have been implicated in swimming behaviour of sperm cells (Fukuda and Touhara, 2005). Given their dynamic regulation in SPGs, we speculate that *Olfr* genes play additional roles in spermatogenesis, other than in sperm physiology. Together with the high number of enriched TF motifs identified at differentially accessible ERVKs and LINE L1 elements, these data highlight a regulatory role for chromatin organization of TEs in SPGs during the transition from development to adult stage (Sundaram et al., 2014; Sundaram and Wysocka, 2020).

Limitations

One limitation of our study is the partial purity of SPGs populations obtained by FACS, due to the use of a purification method that does not fully remove other testicular cells from the samples. This therefore does not exclude the possible influence of contaminating cells on the data and their interpretation. Such influence may explain differences between our datasets and datasets in the literature. Nevertheless, by comparing chromatin landscape in developing and adult SPGs, the study reveals an age-dependent dynamic reorganization of chromatin accessibility in these cells. When integrated with our transcriptomic datasets and published histone PTMs profiles, the data provide novel insight into transcriptome-chromatin dynamics of mouse SPGs from early postnatal life to adulthood with great depth and high resolution. This represents an important resource for future studies on mouse germ cells.

Methods

Mouse husbandry

Male C57Bl/6J mice were purchased from Janvier Laboratories (France) and bred in-house to C57Bl/6J primiparous females to generate males for the experiments. All animals were kept on a reversed 12-h light/12-h dark cycle in a temperature- and humidity-controlled facility with food (M/R Haltung Extrudat, Provimi Kliba SA, Switzerland) and water provided *ad libitum*. Cages were changed once weekly. Animals from 2 independent cohorts generated separately were used for the experiments.

Germ cells isolation

Germ cells were isolated at postnatal day (PND) 8 or 15 for RNA-seq and ATAC-seq, and at postnatal week 20 (adult) for ATAC-seq. Testicular single-cell suspensions were prepared as previously described with slight modifications (Kubota et al., 2004a, 2004b). For PND8 and PND15 cells, testes from 2 animals were pooled for each sample while for adults, testes from individual males were used. Pup testes were collected in sterile HBSS on ice. Tunica albuginea was gently removed from each testis, making sure to keep seminiferous tubules as intact as possible. Tubules were enzymatically digested in 0.25% trypsin-EDTA (ThermoFisher Scientific) and 7mg/ml DNase I (Sigma-Aldrich) solution for 5 min at 37oC. The suspension was vigorously pipetted up and down 10 times and incubated again for 3 min at 37oC. The digestion was stopped by adding 10% fetal bovine serum (ThermoFisher Scientific) and cells were passed through a 20µm-pore-size cell strainer (Miltenyi Biotec) and pelleted by centrifugation at 600g for 7 min at 4oC. Cells were resuspended in PBS-S (PBS with 1% PBS, 10 mM HEPES, 1 mM pyruvate, 1mg/ml glucose, 50 units/ml penicillin and 50 µg/ml streptomycin) and used for sorting. Adult testes were collected and the tunica was removed. Seminiferous tubules were digested in 1 mg/mL collagenase type IV (Sigma Aldrich) and then in 0.25 % Trypsin-EDTA (Gibco), both times for 8-12 minutes and in the presence of DNase I solution (Sigma Aldrich). FBS (Cytiva HyClone) was added to a concentration of 10 % and the cell suspension was filtered through a 40µm-pore-size cell strainer (Corning) and washed with DPBS-S (1 % FBS, 10 mM HEPES (Gibco), 1mM sodium pyruvate (Gibco), 1mg/mL Glucose (Gibco) and 1X Penicillin-Streptomycin (Gibco) in DPBS (Gibco). 5 mL of the single cell suspension was then overlayed on 2 mL 30 % Percoll (Sigma) and centrifuged with disabled break at 600g for 8 minutes at 4 °C. The supernatant was removed and cell suspension was washed twice in DPBS-S and used for sorting.

SPGs enrichment by FACS

For pup testis, dissociated cells were stained with BV421-conjugated anti-β2M, biotin-conjugated anti-THY1 (53-2.1) and PE-conjugated anti-αv-integrin (RMV-7) antibodies. THY1 was detected by staining with Alexa Fluor 488-Sav. For adult testes, cells were stained with anti-α6-integrin (CD49f; GoH3), BV421-conjugated anti-β2 microglobulin (β2M; S19.8) and R-phycoerythrin (PE)-conjugated anti-THY1 (CD90.2; 30H-12) antibodies. α6-Integrin was detected by Alexa Fluor 488-SAv after staining with biotin-conjugated rat anti-mouse IgG1/2a (G28-5) antibody. Prior to FACS, 1 µg/ml propidium iodide (Sigma) was added to cell suspensions to discriminate dead cells. All antibody incubations were performed in PBS-S for at least 30 min at 4oC followed by washing in PBS-S. Antibodies were obtained from BD Biosciences (San Jose, United States) unless otherwise stated. Cell sorting was performed at 4oC on a FACS Aria III 5L using an 85µm nozzle at the Cytometry Facility of University of Zurich. For RNA-seq at PND8 and PND15, cells were collected in 1.5 ml Eppendorf tubes in 500 µL PBS-S, immediately pelleted by centrifugation and snap frozen in liquid N2. Cell pellets were stored at -80oC until RNA extraction. For RNA-seq of adults, 1000 spermatogonial cells (MHC I negative, alpha-6-integrin and Thy1 positive) per male were sorted into PBS. For Omni-ATAC at PND15, 25’000 cells were collected in a separate tube, pelleted by centrifugation and immediately processed using a library preparation protocol (Corces et al., 2017). For Omni-ATAC in adults, 5000 cells from each animal were collected in a separate tube and processed using the same protocol.

Immunocytochemistry

The protocol used for assessing SPGs enrichment after sorting was kindly provided by Jon Oatley at Washington State University, Pullman, USA (Yang et al., 2013). Briefly, 30,000-50,000 cells were adhered to poly-L-Lysine coated coverslips (Corning Life Sciences) in 24-well plates for 1 h. Cells were fixed in freshly prepared 4% PFA for 10 min at room temperature then washed in PBS with 0.1% Triton X-100 (PBS-T). Non-specific antibody binding was blocked by incubation with 10% normal goat serum for 1 h at room temperature. Cells were incubated overnight at 4oC with mouse anti-PLZF (0.2 μg/ml, Active Motif) primary antibody. Alexa488 goat anti-mouse IgG (1 μg/mL, ThermoFisher Scientific) was used for secondary labelling at 4oC for 1 h. Coverslips were washed 3 times, mounted onto glass slides with VectaShield mounting medium containing DAPI (Vector Laboratories) and examined by fluorescence microscopy. SPGs enrichment was determined by counting PLZF+ cells in 10 random fields of view from each coverslip and dividing by the total number of cells present in the same fields of view (DAPI-stained nuclei). The number of PLZF+ and PLZF-cells from the 10 different fields of view was averaged.

RNA extraction and RNA-seq library preparation

For RNA-seq at PND8 and PND15, total RNA was extracted from sorted cells using AllPrep RNA/DNA Micro kit (Qiagen). RNA quality was assessed using a Bioanalyzer 2100 (Agilent Technologies). Samples were quantified using Qubit RNA HS Assay (ThermoFisher Scientific). RNA sequencing libraries were prepared using SMARTer Stranded Total RNA-Seq Kit v3 (Takara Bio USA, Inc) following the recommended protocol with minor adjustments. For PND8 and PND15 SPGs libraries, RNA was fragmented for 4 minutes at 94 °C. After reverse transcription, samples were barcoded using SMARTer Unique Dual Index Kit (Takara Bio USA, Inc). Five PCR cycles were run and PCR products were purified using AMPure XP reagent (Beckman Coulter Life Sciences; bead:sample ratio 0.8X). Ribosomal cDNA was depleted following the protocol and final library amplification was performed with 12 PCR cycles. Libraries were purified twice using AMPure XP reagent (bead:sample ratio 1X) and eluted in Tris buffer. Adult SPGs libraries were prepared using the option to start directly from cells (1000 cells as input). Cells were incubated for 6 minutes at 85 °C and processed as indicated for PND8 and PND15 SPGs. Samples were pooled at equal molarity, as determined in a 100-800 bp window on the Bioanalyzer.

Omni-ATAC library preparation and sequencing

Chromatin accessibility was profiled from PND15 and adult SPGs. Libraries were prepared according to Omni-ATAC protocol, starting from 25,000 PND15 and 5000 adult sorted SPGs (Corces et al., 2017). Briefly, sorted cells were lysed in cold lysis buffer (10 mM Tris-HCl pH 7.4, 10 mM NaCl, 3 mM MgCl2, 0.1% NP40, 0.1% Tween-20, and 0.01% digitonin) and nuclei were pelleted and transposed using Nextera Tn5 (Illumina) for 30 min at 37oC in a thermomixer with shaking at 1000 rpm. Transposed fragments were purified using MinElute Reaction Cleanup Kit (Qiagen). Following purification, libraries were generated by PCR amplification using NEBNext High-Fidelity 2X PCR Master Mix (New England Biolabs) and purified using Agencourt AMPure XP magnetic beads (Beckman Coulter) to remove primer dimers (78bp) and fragments of 1000-10,000 bp. Library quality was assessed on an Agilent High Sensitivity DNA chip using the Bioanalyzer 2100 (Agilent Technologies). 6 samples were sequenced from PND15 and 5 from adult SPGs.

RNA-seq analysis

Quality control and alignment: 100bp single end sequencing was performed at the Functional Genomics Center Zurich (FGCZ) using a Novaseq SP flowcell on the Novaseq 6000 platform. Quality assessment of FASTQ files was done using FastQC (Andrews et al., 2012) (version 0.11.8). TrimGalore (Krueger, 2015) (version 0.6.2) was used to trim adapters and low-quality ends from reads with Phred score less than 30 (-q 30) and to discard trimmed reads shorter than 30bp (--length 30). Trimmed reads were pseudo-aligned using Salmon (Patro et al., 2017) (version 0.9.1) with automatic detection of the library type (-l A), correcting for sequence-specific bias (--seqBias) and correcting for fragment GC bias correction (--gcBias) on a transcript index prepared for the Mouse genome (GRCm38) from GENCODE (version M18) (Harrow et al., 2012), with additional piRNA precursors and TEs (concatenated by family) from Repeat Masker as in (Gapp et al., 2018).

Downstream analysis

Data analyses and plotting were conducted with R (R Core Team, 2020) (version 3.6.2) using packages from The Comprehensive R Archive Network (CRAN) (*https://cran.r-project.org*) and Bioconductor (Huber et al., 2015). Pre-filtering of genes was done using the filterByExpr function from edgeR (Robinson et al., 2009) (version 3.28.1) with a design matrix requiring at least 15 counts (min.counts=15). Normalization factors were obtained using TMM normalization (Robinson and Oshlack, 2010) from edgeR package and differential gene expression analyses were done using limma-voom (Law et al., 2014) pipeline from limma (Ritchie et al., 2015) (version 3.42.2). GO enrichment analyses were performed using g:Profiler (*https://biit.cs.ut.ee/gprofiler/gost*) querying against Mus musculus database.

Deconvolution analyses

Hermann et al., 2018: Raw data were obtained from PND6 and adult sorted ID4-EGFP+ SPGs. Using the cluster IDs provided in the supplemental datasets, CIBERSORTx analysis (Newman et al., 2019) was performed to estimate the proportion of labeled cell types in each RNA-seq library using the default settings (*G.min*=300, *G.max*=500, q=0.1). For deconvolution analyses, we used the aggregated P6 and adult datasets as references.

Tan et al., 2020: Raw data from E18, P2 and P7 were obtained. We applied the reported filtering and normalization parameters, followed by UMAP analysis (Becht et al., 2018) using the top 10 principal components (PCs). For clustering, the Seurat package was employed (Butler et al., 2018, *dims*=1:10 in *FindNeighbors*, *resolution*=0.025 in *FindClusters*) and resultant cell counts in each cluster corresponded to those reported in Tan et al., 2020. Clusters identified as germ, Sertoli, stroma, Leydig and peritubular myoid (PTM) cells were validated using specific marker genes as per Tan et al., 2020; Green et al., 2018; and Hermann et al., 2018. Cells not classified into these clusters were categorized as 'Other Cells.' Then, CIBERSORTx was utilized to estimate the proportion of these labeled cell types in each RNA-seq library using the default settings (*G.min*=300, *G.max*=500, *q*=0.1). For the deconvolution analyses, two replicate libraries from PND7 were used as references, and their averaged results were reported. Additionally, we performed a re-analysis focusing on re-clustered germ cells. Within this subset, SSCs and differentiating SPGs were identified, using marker genes referenced in Tan et al., 2020; Green et al., 2018; and Hermann et al., 2018.

Green et al., 2018: Raw data from adult were obtained. We applied the reported filtering and normalization parameters, followed by UMAP analysis (Becht et al., 2018) using the top 10 principal components (PCs). For clustering, the Seurat package (Butler et al., 2018, *dims*=1:20 in *FindNeighbors, resolution*=0.011 in *FindClusters*). Clusters identified as germ, Sertoli, stroma, Leydig, and peritubular myoid (PTM) cells were validated using specific marker genes as per Tan et al., 2020; Green et al., 2018; and Hermann et al., 2018. Cells not classified into these clusters were categorized as 'Other Cells.' Then CIBERSORTx was utilized to estimate the proportion of these labeled cell types in each RNA-seq library using the default settings (*G.min*=300, *G.max*=500, *q*=0.1). For deconvolution analyses, we used 8 seminiferous tubule (ST datasets), 3 spermatogonia enriched (SPGs datasets) and 8 Sertoli enriched (SER datasets) as our references separately, and the averages of each group were reported.

ATAC-seq analysis

Quality control, alignment, and peak calling: Paired-end (PE) sequencing was performed on PND15 and adult SPGs on Illumina HiSeq2500 platform at FGCZ. FASTQ files were assessed for quality using FastQC (Andrews et al., 2012) (version 0.11.8). QC was performed using TrimGalore (Krueger, 2015) (version 0.6.2) in PE mode (--paired), trimming adapters, low-quality ends (-q 30) and discarding reads shorter than 30bp after trimming (--length 30). Alignment on GRCm38 genome was performed using Bowtie2 (Langmead and Salzberg, 2012) (version 2.3.5) with the following parameters: allowing fragments up to 2kb to align (-X 2000), entire read alignment (--end-to-end), suppressing unpaired alignments for paired reads (--no-mixed), suppressing discordant alignments for paired reads (--no-discordant) and minimum acceptable alignment score with respect to read length (--score-min L,- 0.4,-0.4). Using alignmentSieve (version 3.3.1) from deepTools (Ramirez et al., 2016) (version 3.4.3), aligned data (BAM files) were adjusted for read start sites to represent the center of the transposon cutting event (--ATACshift), and filtered for reads with a high mapping quality (--minMappingQuality 30). Reads mapping to the mitochondrial chromosome and ENCODE blacklisted regions were filtered out. To call nucleosome-free regions, all aligned files were merged within groups (PND15 and adult), sorted and indexed using SAMtools (Li et al., 2009) (version 0.1.19) and nucleosome-free fragments (NFFs) were obtained by selecting alignments with a template length between 40 and 140bp in length. Peak calling on NFFs was performed using MACS2 (Zhang et al., 2008) (version 2.2.7.1) with mouse genome size (-g 2744254612) and PE BAM file format (-f BAMPE).

Differential accessibility analysis

These analyses were conducted in R (version 3.6.2) using packages from CRAN (*https://cran.r-project.org*) and Bioconductor (Huber et al., 2015). Peaks were annotated based on overlap with GENCODE (version M18) (Harrow et al., 2012) transcript and/or distance to the nearest TSS (*https://github.com/mansuylab/SC_postnatal_adult/bin/annoPeaks.R*). The number of extended reads overlapping in peak regions was calculated using csaw package (Lun and Smyth, 2015) (version 1.20.0). Peak regions that did not have at least 15 reads in at least 40% of the samples were filtered out. Normalization factors were obtained on filtered peak regions using TMM normalization method (Robinson and Oshlack, 2010) and differential analysis on peaks (PND15 versus adult) was performed using Genewise Negative Binomial Generalized Linear Models with Quasi-likelihood (glmQLFit) Tests from edgeR package (Robinson et al., 2009) (version 3.28.1). Peak regions with an absLog2FC≥1 and adjusted-P≤0.05 were categorized as DARs.

Downstream analysis

Heatmaps of normalized ATAC-seq signal were created using deepTools with default parameters. Genomic annotation of ATAC-seq peak and DARs and identification of closest gene DARs were performed using ChIPpeakAnno (Zhu et al., 2010). GO enrichment analysis of genes associated to DARs was performed using g:Profiler (Raudvere et al., 2019) querying against the *Mus musculus* database. For the epigenomic annotation of DARs, publicly available signal files and peak files for all histone marks were used (Cheng et al., 2020). Heatmaps of ChIP-seq signal for histone PTMs around ATAC-seq peaks and DARs were generated with deepTools. Overlap between ChIP-seq peaks and DARs was generated using Intervene (Khan et al., 2017). TF motif analysis was performed using MEME-ChIP (Ma et al., 2014) and motif identification was done querying identified motifs against JASPAR database (Castro-Mondragon et al., 2021). Quantification and identification of differences in chromatin accessibility between promoter regions of DEGs was performed with deepStats (Richard 2019). To quantify differences in chromatin accessibility between DEGs, bamscale cov (Pongor et al., 2020) was employed using ATAC-seq BAM files as input and genomic coordinates of up-regulated and down-regulated genes. Kruskal-Wallis test was used to test for significant differences in chromatin accessibility between each category of DEGs at postnatal and adult stage.

Differential accessibility analysis at TEs

TE gene transfer format (GTF) file was obtained from *http://labshare.cshl.edu/shares/mhammelllab/www-data/TEtranscripts/TE_GTF/mm10_rmsk_TE.gtf.gz* on 03.02.2020. This file provides hierarchical information about TEs: Class (level 1, eg. LTR), family (level 2, eg. LTR ->L1), subtype (level 3, eg. LTR-> L1->L1_Rod), and locus (level 4, eg. LTR->L1 -> L1_Rod -> L1_Rod_dup1). TE loci were annotated based on overlap with GENCODE (version M18) as described above for ATAC-seq peaks. Filtered BAM files (without reads mapping to blacklisted or mitochondrial regions) were used to analyze TEs. Mapped reads were assigned to TEs using featureCounts from R package Rsubread (Liao et al., 2019) (version 2.0.1) and were summarized to Subtypes (level 3), allowing for multi-overlap with fractional counts, while ignoring duplicates. The number of extended reads overlapping at TE loci were obtained using csaw package (Lun and Smyth, 2015) (version 1.20.0). Subtypes without at least 15 reads, and loci without at least 5 reads in at least 40% of samples were filtered out. Normalization and differential accessibility analysis were performed as described above. Subtypes which had an absolute Log2FC≥0.5 and adjusted-P≤0.05 were categorized as differentially accessible subtypes and loci with an absLog2FC≥1 and adjusted-P≤0.05 were categorized as differentially accessible loci. For further downstream data analysis, only differentially accessible loci or differentially accessible subtypes were considered. TF motif enrichment analysis was performed using the marge package (Amezquita, 2018) (version 0.0.4.9999), which is a wrapper around the Homer tool (Heinz et al., 2010) (version 4.11.1).

Authors contribution

ILC and IMM designed and conceived the study. ILC performed all RNA-seq, ICC and ATAC-seq experiments with support of LCS. DKT and RGA-M analyzed the RNA-seq, ATAC-seq, ChIP-seq with significant support from PLG. KU performed deconvolution analysis. OUF and MC performed chromatin accessibility analysis of DEGs. ILC, DKT and RGA-M prepared figures. RGA-M took care of all revisions, prepared Supp. Fig. 2 for the revised manuscript and all figures for the response to reviewers. ILC and RGA-M interpreted the data with significant input from DKT, PLG, KU and IMM. ILC, DKT, RGA-M, and IMM wrote the manuscript, with significant input from PLG. All authors read and accepted the submitted version of the manuscript.

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Competing interest

The authors declare that they have no competing interests.

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Data and material availability

Sequencing data is available via the EBI BioStudies database within the ArrayExpress collection (RNA-seq: E-MTAB-12721; ATAC-seq: E-MTAB-12722). The code employed for data analysis is available from *https://github.com/mansuylab/SC_postnatal_adult/*. Additional information is available upon request.

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Editors

Reviewing Editor
Wei Yan
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Senior Editor
Wei Yan
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Reviewer #1 (Public Review):

Summary

This study was designed to investigate changes in gene expression and associated chromatin accessibility patterns in spermatogonia in mice at different postnatal stages from pups to adults. The objective was to describe dynamic changes in these patterns that potentially correlate with functional changes in spermatogonia as a function of development and reproductive maturation. The potential utility of this information is to serve as a reference against which similar data from animals subjected to various disruptive environmental influences can be compared.

Major Strengths and Weaknesses of the Methods and Results

A strength of the study is that it reviews previously published datasets describing gene expression and chromatin accessibility patterns in mouse spermatogonia. A weakness of the study is that it is not clear what new information is provided by the data provided that was not already known from previously published studies (see below). Specific weaknesses include the following...

- Terminology - In the Abstract and first part of the Introduction the authors use the generic term "spermatogonial cells" in a manner that seems to be referring primarily to spermatogonial stem cells (SSCs) but initially ignores the well-known heterogeneity among spermatogonia - particularly the fact that only a small proportion of developing spermatogonia become SSCs - and ONLY those SSCs and NOT other developing spermatogonia - support steady-state spermatogenesis by retaining the capacity to either self-renew or contribute to the differentiating spermatogenic lineage throughout the male reproductive lifespan. The authors eventually mention other types of developing male germ cells, but their description of prospermatogonial stages that precede spermatogonial stages is deficient in that M-prospermatogonia - which occur after PGCs but before T1-prospermatogonia - are not mentioned. This description also seems to imply that all T2-prospermatogonia give rise to SSCs which is far from the case. It is the case that prospermatogonia give rise to spermatogonia, but only a very small proportion of undifferentiated spermatogonia form the foundational SSCs and ONLY SSCs possess the capacity to either self-renew or give rise to sequential waves of spermatogenesis.

- Introduction - Statements regarding distinguishing transcriptional signatures in spermatogonia at different postnatal stages appear to refer to ALL subtypes of spermatogonia present at each stage collectively, thereby ignoring the well-known fact that there are distinct spermatogonial subtypes present at each postnatal stage and that some of those occur at certain stages but not at others. This brings into question the usefulness of the authors' discussion of what types of genes are expressed and/or what types of changes in chromatin accessibility are detected in spermatogonia at each stage.

- Methodology - The authors based recovery (enrichment) of spermatogonia from male pups on FACS sorting for THY1 and RMV-1. While sorting total testis cells for THY1+ cells does enrich for spermaogonia, this approach is now known to not be highly specific for spermatogonia (somatic cells are also recovered) and definitely not for SSCs. There are more effective means for isolating SSCs from total testis cells that have been validated by transplantation experiments (e.g. use of the Id4/eGFP transgene marker).

The authors then used "deconvolution" of bulk RNA-seq data in an attempt to discern spermatogonial subtype-specific transcriptomes. It is not clear why this is necessary or how it is beneficial given the availability of multiple single-cell RNA-seq datasets already published that accomplish this objective quite nicely - as the authors essentially acknowledge. Beyond this concern, a potential flaw with the deconvolution of bulk RNA-seq data is that this is a derivative approach that requires assumptions/computational manipulations of apparent mRNA abundance estimates that may confound interpretation of the relative abundance of different cellular subtypes within the hetergeneous cell population from which the bulk RNA-seq data is derived. Bottom line, it is not clear that this approach affords any experimental advantage over use of the publicly available scRNA-seq datasets and it is possible that attempts to employ this approach may be flawed yielding misleading data.

- Results & Discussion - In general, much of the information reported in this study is not novel. The authors' discussion of the makeup of various spermatogonial subtypes in the testis at various ages does not really add anything to what has been known for many years on the basis of classic morphological studies. Further, as noted above, the gene expression data provided by the authors on the basis of their deconvolution of bulk RNA-seq data does not add any novel information to what has been shown in recent years by multiple elegant scRNA-seq studies - and, in fact, as also noted above - represents an approach fraught with potential for misleading results. The potential value of the authors' report of "other cell types" not corresponding to major somatic cell types identified in earlier published studies seems quite limited given that they provide no follow-up data that might indicate the nature of these alternative cell types. Beyond this, much of the gene expression and chromatin accessibility data reported by the authors - by their own admission given the references they cite - is largely confirmatory of previously published results. Similarly, results of the authors' analyses of putative factor binding sites within regions of differentially accessible chromatin also appear to confirm previously reported results. Ultimately, it is not at all novel to note that changes in gene expression patterns are accompanied by changes in patterns of chromatin accessibility in either related promoters or enhancers. The discussion of these observations provided by the authors takes on more of a review nature than that of any sort of truly novel results. As a result, it is difficult to discern how the data reported in this manuscript advance the field in any sort of novel or useful way beyond providing a review of previously published studies on these topics.

Likely impact - The likely impact of this work is relatively low because, other than the value it provides as a review of previously published datasets, the new datasets provided are not novel and so do not advance the field in any significant manner.

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Reviewer #2 (Public Review):

This revised manuscript attempts to explore the underlying chromatin accessibility landscape of spermatogonia from the developing and adult mouse testis. The key criticism of the first version of this manuscript was that bulk preparations of mixed populations of spermatogonia were used to generate the data that form the basis of the entire manuscript. To address this concern, the authors applied a deconvolution strategy (CIBERSORTx (Newman et al., 2019)) in an attempt to demonstrate that their multi-parameter FACS isolation (from Kubota 2004) of spermatogonia enriched for PLZF+ cells recovered spermatogonial stem cells (SSCs). PLZF (ZBTB16) protein is a transcription factor known to mark all or nearly all undifferentiated spermatogonia and some differentiating spermatogonia (KIT+ at the protein level) - see Niedenberger et al., 2015 (PMID: 25737569). The authors' deconvolution using single-cell transcriptomes produced at postnatal day 6 (P6) argue that 99% of the PLZF+ spermatogonia at P8 are SSCs, 85% at P15 and 93% in adults. Quite frankly given the established overlap between PLZF and KIT and known identity of spermatogonia at these developmental stages, this is impossible. Indeed - the authors' own analysis of the reference dataset demonstrates abundant PLZF mRNA in P6 progenitor spermatogonia - what is the authors' explanation for this observation? The same is essentially true in the use of adult references for celltype assignment. The authors found 63-82% of SSCs using this different definition of types (from a different dataset), begging the question of which of these results is true.

In their rebuttal, the authors also raise a fair point about the precision of differential gene expression among spermatogonial subsets. At the mRNA level, Kit is definitely detectable in undifferentiated spermatogonia, but it is never observed at the protein level until progenitors respond to retinoic acid (see Hermann et al., 2015). I agree with the authors that the mRNAs for "cell type markers" are rarely differentially abundant at absolute levels (0 or 1), but instead, there are a multitude of shades of grey in mRNA abundance that "separate" cell types, particularly in the male germline and among the highly related spermatogonial subtypes of interest (SSCs, progenitor spermatogonia and differentiating spermatogonia). That is, spermatogonial biology should be considered as a continuous variable (not categorical), so examining specific cell populations with defined phenotypes (markers, function) likely oversimplifies the underlying heterogeneity in the male germ lineage. But, here, the authors have ignored this heterogeneity entirely by selecting complex populations and examining them in aggregate. We already know that PLZF protein marks a wide range of spermatogonia, complicating the interpretation of aggregate results emerging from such samples. In their rebuttal, the authors nicely demonstrate the existence of these mixtures using deconvolution estimation. What remains a mystery is why the authors did not choose to perform single-cell multiome (RNA-seq + ATAC-seq) to validate their results and provide high-confidence outcomes. This is an accessible technique and was requested after the initial version, but essentially ignored by the authors.

A separate question is whether these data are novel. A prior publication by the Griswold lab (Schleif et al., 2023; PMID: 36983846) already performed ATAC-seq (and prior data exist for RNA-seq) from germ cells isolated from synchronized testes. These existing data are higher resolution than those provided in the current manuscript because they examine germ cells before and after RA-induced differentiation, which the authors do not base on their selection methods. Another prior publication from the Namekawa lab extensively examined the transcriptome and epigenome in adult testes (Maezawa et al., 2000; PMID: 32895557; and several prior papers). The authors should explain how their results extend our knowledge of spermatogonial biology in light of the preceding reports.

The authors are also encouraged to improve their use of terminology to describe the samples of interest. The mitotic male germ cells in the testis are called spermatogonia (not spermatogonial cells, because spermatogonia are cells). Spermatogonia arise from Prospermatogonia. Spermatogonia are divisible into two broad groups: undifferentiated spermatogonia (comprised of few spermatogonial stem cells or SSCs and many more progenitor spermatogonia - at roughly 1:10 ratio) and differentiating spermatogonia that have responded to RA. The authors also improperly indicate that SSCs directly produce differentiating spermatogonia - indeed, SSCs produce transit-amplifying progenitor spermatogonia, which subsequently differentiate in response to retinoic acid stimulation. Further, the use of Spermatogonial cells (and SPGs) is imprecise because these terms do not indicate which spermatogonia are in question. Moreover, there have been studies in the literature which have used similar terms inappropriately to refer to SSCs, including in culture. A correct description of the lineage and disambiguation by careful definition and rigorous cell type identification would benefit the reader.

Overall, my concern from the initial version of this manuscript stands - critical methodological flaws prevent interpretation of the results and the data are not novel. Readers should take note that results in essentially all Figures do not reflect the biology of any one type of spermatogonium.

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Reviewer #3 (Public Review):

In this study, Lazar-Contes and colleagues aimed to determine whether chromatin accessibility changes in the spermatogonial population during different phases postnatal mammalian testis development. Because actions of the spermatogonial population set the foundation for continual and robust spermatogenesis and the gene networks regulating their biology are undefined, the goal of the study has merit. To advance knowledge, the authors used mice as a model and isolated spermatogonia from three different postnatal developmental age points using cell sorting methodology that was based on cell surface markers reported in previous studies and then performed bulk RNA-sequencing and ATAC-sequencing. Overall, the technical aspects of the sequencing analyses and computational/bioinformatics seems sound but there are several concerns with the cell population isolated from testes and lack of acknowledgement for previous studies that have also performed ATAC-sequencing on spermatogonia of mouse and human testes. The limitations, described below, call into question validity of the interpretations and reduce the potential merit of the findings.

I suggest changing the acronym for spermatogonial cells from SC to SPG for two reasons. First, SPG is the commonly used acronym in the field of mammalian spermatogenesis.

Second, SC is commonly used for Sertoli Cells.

The authors should provide a rationale for why they used postnatal day 8 and 15 mice.

The FACS sorting approach used was based on cell surface proteins that are not germline specific so there was undoubtedly somatic cells in the samples used for both RNA and ATAC sequencing. Thus, it is essential to demonstrate the level of both germ cell and undifferentiated spermatogonial enrichment in the isolated and profiled cell populations. To achieve this, the authors used PLZF as a biomarker of undifferentiated spermatogonia. Although PLZF is indeed expressed by undifferentiated spermatogonia, there have been several studies demonstrating that expression extends into differentiating spermatogonia. In addition, PLZF is not germ cell specific and single cell RNA-seq analyses of testicular tissue has revealed that there are somatic cell populations that express Plzf, at least at the mRNA level. For these reasons, I suggest that the authors assess the isolated cell populations using a germ cell specific biomarker such as DDX4 in combination with PLZF to get a more accurate assessment of the undifferentiated spermatogonial composition. This assessment is essential for interpretation of the RNA-seq and ATAC-seq data that was generated.

A previous study by the Namekawa lab (PMID: 29126117) performed ATAC-seq on a similar cell population (THY1+ FACS sorted) that was isolated from pre-pubertal mouse testes. It was surprising to not see this study referenced to in the current manuscript. In addition, it seems prudent to cross-reference the two ATAC-seq datasets for commonalities and differences. In addition, there are several published studies on scATAC-seq of human spermatogonia that might be of interest to cross-reference with the ATAC-seq data presented in the current study to provide an understanding of translational merit for the findings.

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Author response:

The following is the authors’ response to the original reviews.

> ***Reviewer #1:***
>
> *Weaknesses:*
>
> *There appears to be a lack of basic knowledge of the process of spermatogenesis. For instance, the statement that "During the first week of postnatal life, a population of SCs continues to proliferate to give rise to undifferentiated Asingle (As), Apaired (Apr) and Aaligned (Aal) cells. The remaining SCs differentiate to form chains of daughter cells that become primary and secondary spermatocytes around postnatal day (PND) 10 to 12." is inaccurate. The Aal cells are the spermatogonial chains, the two are not distinct from one another. In addition, the authors fail to mention spermatogonial stem cells which form the basis for steady-state spermatogenesis. The authors also do not acknowledge the well-known fact that, in the mouse, the first wave of spermatogenesis is distinct from subsequent waves. Finally, the authors do not mention the presence of both undifferentiated spermatogonia (aka - type A) and differentiating spermatogonia (aka - type B). The premise for the study they present appears to be the implication that little is known about the dynamics of chromatin during the development of spermatogonia. However, there are published studies on this topic that have already provided much of the information that is presented in the current manuscript.*

Regarding the inaccuracy and incompleteness of some of the statements about spermatogonial cells and spermatogenesis. In the Introduction, we replaced the following statement: "During the first week of postnatal life, a population of SCs continues to proliferate to give rise to undifferentiated Asingle (As), Apaired (Apr) and Aaligned (Aal) cells. The remaining SCs differentiate to form chains of daughter cells that become primary and secondary spermatocytes around postnatal day (PND) 10 to 12." by: “Spermatogonial cells (SPGs) are the initiators and supporting cellular foundation of spermatogenesis in testis in many species, including mammals. In the mammalian testis, the founding germ cells are primordial germ cells (PGCs), which give rise sequentially to different populations of SPGs : primary transitional (T1)-prospermatogonia (ProSG), secondary transitional (T2)-ProSG, and then spermatogonial stem cells (SSCs) (McCarrey, 2013; Rabbani et al., 2022; Tan et al., 2020). The ProSG population is exhausted by postnatal day (PND) 5 (Drumond et al., 2011) and by PND6-8, distinct SPGs subtypes can be distinguished on the basis of specific marker proteins and regenerative capacity (Cheng et al., 2020; Ernst et al., 2019; Green et al., 2018; Hermann et al., 2018; Tan et al., 2020).

SSCs represent an undifferentiated population of SPGs that retain regenerative capacity and divide to either self-renew or generate progenitors that initiate spermatogenic differentiation, giving rise to differentiating SPGs (diff-SPGs). Diff-SPGs form chains of daughter cells that become primary and secondary spermatocytes around PND10 to 12. Spermatocytes then undergo meiosis and give rise to haploid spermatids that develop into spermatozoa. Spermatozoa are then released into the lumen of seminiferous tubules and continue to mature in the epididymis until becoming capable of fertilization by PND42-48 in mice (Kubota and Brinster, 2018; Rooij, 2017).”

Regarding the premise and implications of our findings. We clarified the premise of our finding in the revised manuscript. The following statement was included in the Discussion: "our findings complement existing datasets on spermatogonial cells by providing parallel transcriptomic and chromatin accessibility maps at high resolution from the same cell populations at early postnatal, late postnatal and adult stages collected from single individuals (for adults)".

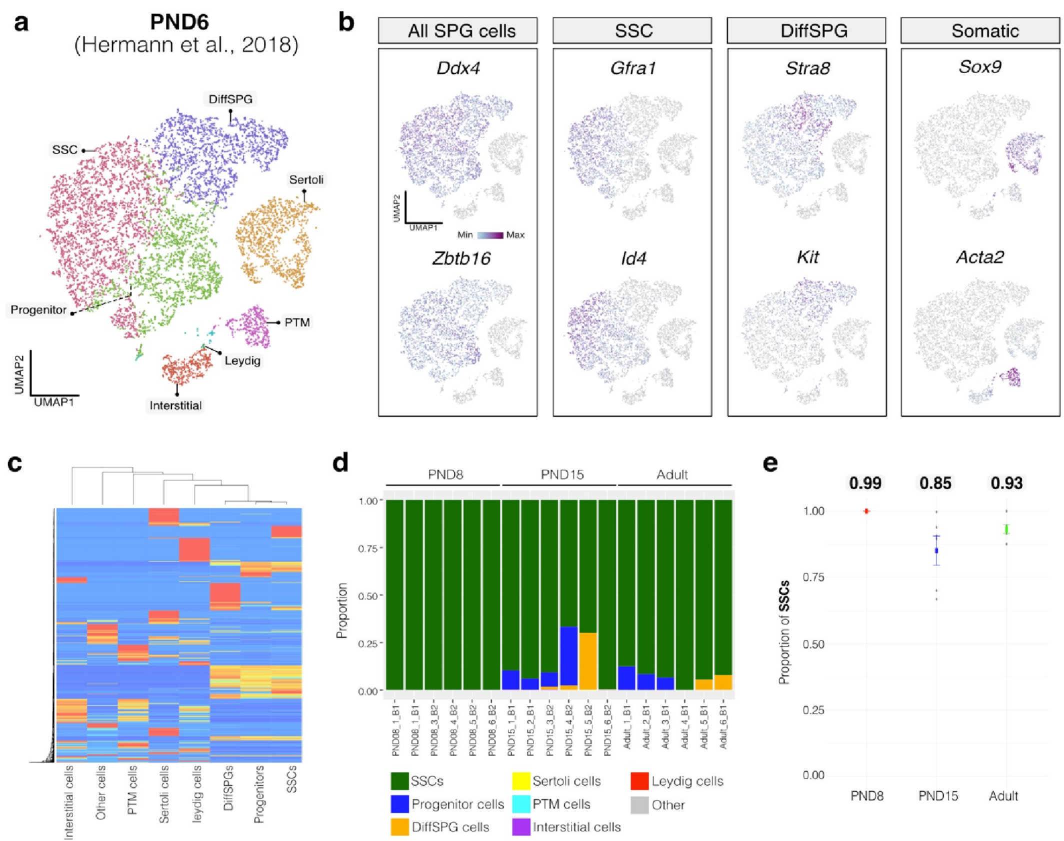
> *It is not clear which spermatogonial subtype the authors intended to profile with their analyses. On the one hand, they used PLZF to FACS sort cells. This typically enriches for undifferentiated spermatogonia. On the other hand, they report detection in the sorted population of markers such as c-KIT which is a well-known marker of differentiating spermatogonia, and that is in the same population in which ID4, a well-known marker of spermatogonial stem cells, was detected. The authors cite multiple previously published studies of gene expression during spermatogenesis, including studies of gene expression in spermatogonia. It is not at all clear what the authors' data adds to the previously available data on this subject.*
>
> *The authors analyzed cells recovered at PND 8 and 15 and compared those to cells recovered from the adult testis. The PND 8 and 15 cells would be from the initial wave of spermatogenesis whereas those from the adult testis would represent steady-state spermatogenesis. However, as noted above, there appears to be a lack of awareness of the well-established differences between spermatogenesis occurring at each of these stages.*

We applied computational deconvolution to our bulk RNA-seq datasets, employing publicly available single-cell RNA-seq datasets, to estimate and identify cellular composition. Trained on high-quality RNA-seq datasets from pure or single-cell populations, deconvolution algorithms create expression matrices reflecting the cellular diversity in reference datasets. These cell-type-specific expression matrices are subsequently used to determine the cellular composition of bulk RNA-seq samples with unknown cellular components (Cobos et al., 2023).

For our analysis, we chose CIBERSORTx (Newman et al., 2019), recognized as the most advanced deconvolution algorithm to date, employing it with three high-quality, publicly available single-cell RNA-seq datasets. First, we assessed the cellular composition of all our RNA-seq libraries, using datasets generated by (Hermann et al., 2018) which characterized the single-cell transcriptomes of testicular cells and various populations of spermatogonial progenitor cells (SPGs) in early postnatal (PND6) and adult stages. This enabled us to not only address potential somatic cell contamination but also to analyse the composition of isolated SPGs using a unified dataset source.

Author response image 1.

Deconvolution analysis of bulk RNA-seq samples using PND6 single-cell RNA seq from Hermann et al, 2018 **a**. Seurat clusters from PND6 single-cell RNA-seq. **b.** Feature maps of gene expression for markers of SPGs and somatic cells. **c.** Gene expression signature matrix from PND6 single-cell RNA-seq datasets. d. Barplot of estimated cellular proportions for all bulk RNA-seq libraries reported in this study. **e.** Dotplot of the average estimated proportion of SSCs in all bulk RNA-seq libraries reported in this study.

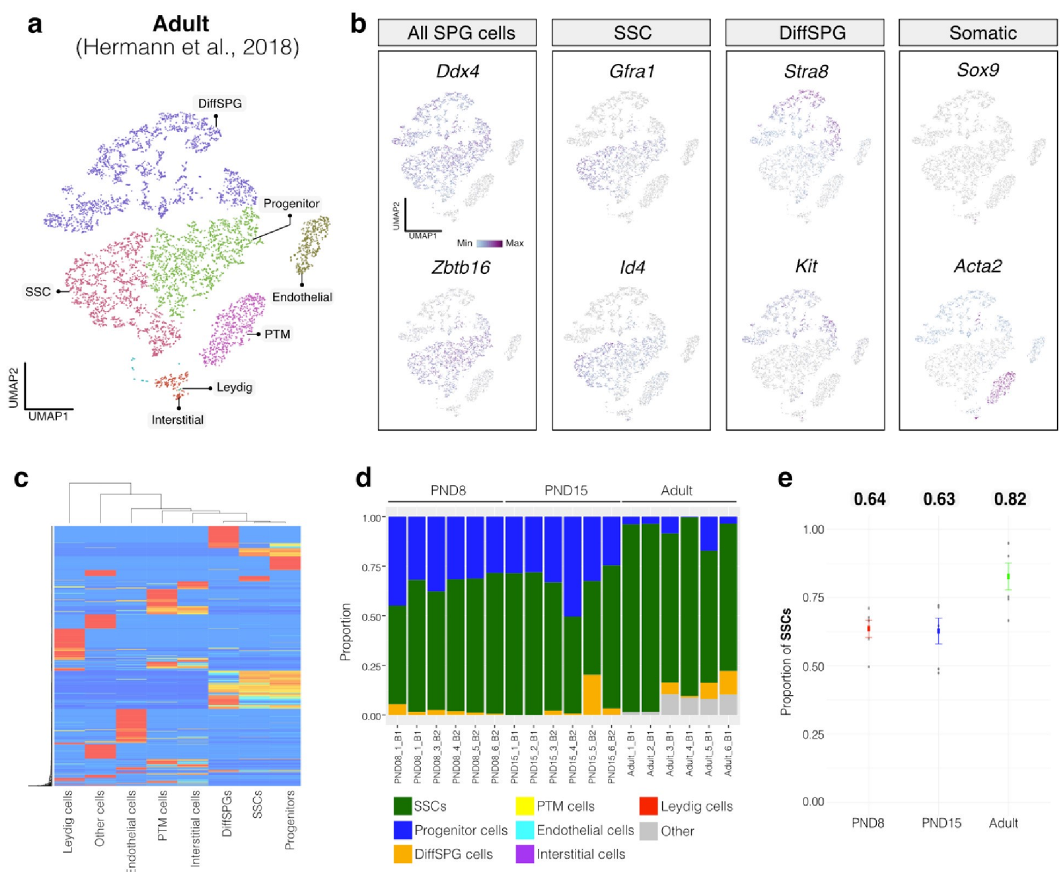


By re-analyzing the single-cell RNA-seq datasets, we identified distinct cell-type clusters, marked by specific cellular markers as reported in the original and subsequent studies (Author response image 1a,b and Author response image 2a,b). Then, CIBERSORTx generated gene-expression signature matrices and estimated the cell-type proportions within our 18 bulk RNA-seq libraries. Evaluation of our postnatal libraries (PND8 and 15) against a PND6 signature matrix revealed a predominant derivation from SPGs, with average estimated proportions of spermatogonial stem cells (SSCs) being 0.99 and 0.85 for PND8 and PND15 samples, respectively (Author response image 1c-e). Notably, the analysis of PND15 libraries also suggested the presence of additional SPGs types, including progenitors and differentiating SPGs (Author response image 1d), albeit at lower frequency.

Similarly, evaluation of our adult RNA-seq libraries, using an adult signature matrix, showed an average SSC proportion of 0.82, indicating a primary derivation from SSC cells. Consistent with the findings from PND15 libraries, our deconvolution analysis also suggests the presence of additional SPG types, including progenitors and differentiating SPGs (Author response image 1d). However, unlike our early and late postnatal stage libraries, the deconvolution analysis of adult libraries indicated the presence of other cell types (labeled "Other"), not corresponding to the major somatic cell types identified by Hermann et al. 2018. The estimated average proportion of these cells was less than 0.05 in two adult libraries and 0.10 in the others. This variance in cellular composition underlines the deconvolution method's effectiveness in dissecting complex cellular compositions in bulk RNA-seq samples.

Author response image 2.

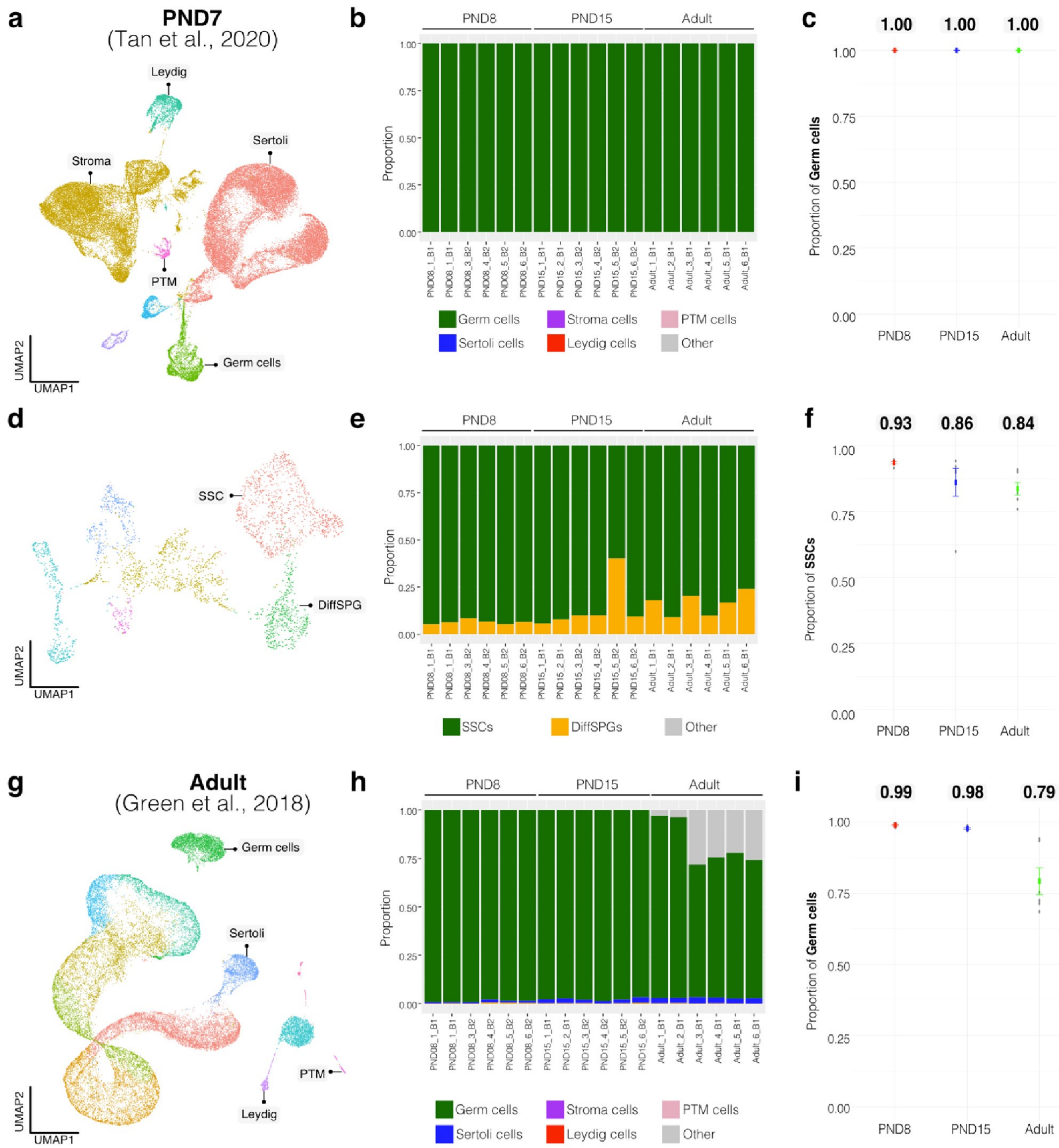
Deconvolution analysis of bulk RNA-seq samples using Adult single-cell RNA seq (Hermann et al, 2018) **a.** Seurat clusters from Adult single-cell RNA-seq. **b.** Feature maps of gene expression for markers of SPG and somatic cells. **c.** Gene expression signature matrix from Adult single-cell RNA-seq datasets. d. Barplot of estimated cellular proportions for all bulk RNA-seq libraries reported in this study. **e.** Dotplot of the average estimated proportion of SSCs in all bulk RNA-seq libraries reported in this study.



To further validate our observations, we re-analyzed two additional testicular single-cell RNA-seq datasets derived from an early postnatal stage (PND7) (Tan et al., 2020) and adult (Green et al., 2018) (Author response image 3a,b and Author response image 4a,b). We identified distinct cell-type clusters, marked by specific cellular markers (Author response image 3a,b and Author response image 4a,b), and proceeded with the deconvolution analysis using CIBERSORTx. Evaluation of our postnatal libraries (PND8 and 15) against the PND7 signature matrix from Tan et al., 2020 confirmed a derivation from germ cells (Author response image 3d,e), in particular from SSCs (Author response image 3g), with average estimated proportions of SSCs being 0.93 and 0.86 for PND8 and PND15 samples, respectively, and the rest estimated to be in origin from differentiating SPGs (Author response image 3g,h). In the case of the adult samples, evaluation against the adult signature matrix from Green et al., 2018 confirmed a predominant derivation from SSCs, with average estimated proportions of SSCs being 0.79, consistent with the 0.82 estimated proportion from Hermann et al., 2018.

Author response image 3.

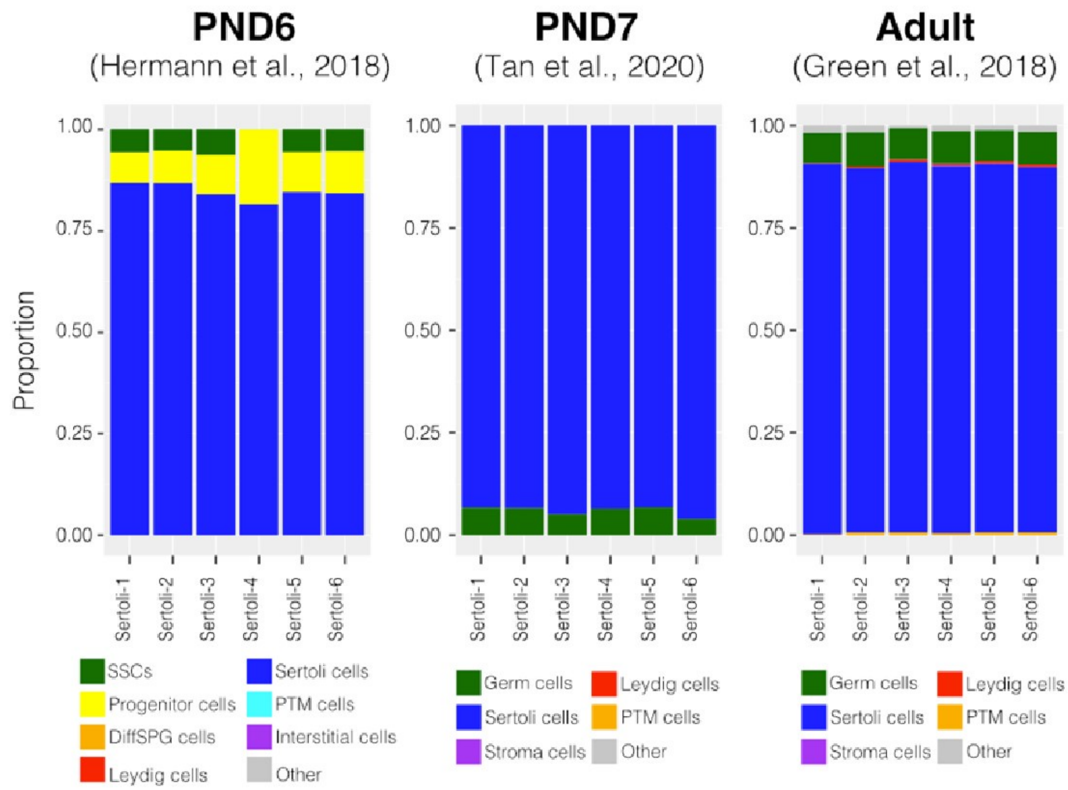
Deconvolution analysis of bulk RNA-seq samples with additional single-cell datasets. Seurat clusters from PND7 single-cell RNA-seq (Tang 2020). **b.** Barplot of estimated cellular proportions for all bulk RNA-seq libraries reported in this study. **c.** Dotplot of the average estimated proportion of germ cells in all bulk RNA-seq libraries reported in this study. **d.** Re-clustering of germ cell cluster shown in a. **e.** Barplot of estimated cellular proportions for all bulk RNA-seq libraries reported in this study. **f.** Dotplot of the average estimated proportion of SSCs in all bulk RNA-seq libraries reported in this study. **g.** Seurat clusters from adult single-cell RNA-seq (Green et al., 2018). h**.** Barplot of estimated cellular proportions for all bulk RNA-seq libraries reported in this study. **i.** Dotplot of the average estimated proportion of germ cells in all bulk RNA-seq libraries reported in this study.



To further validate our deconvolution strategy, we interrogated the cellular composition of bulk RNA-seq libraries derived from cellular populations enriched in Sertoli cells, generated by our group using a similar enrichment/sorting strategy (Thumfart et al., 2022). As expected, our results show that all our libraries are mainly composed of Sertoli cells suggesting that the deconvolution strategy employed is accurate in detecting cell-type composition (Author response image 4).

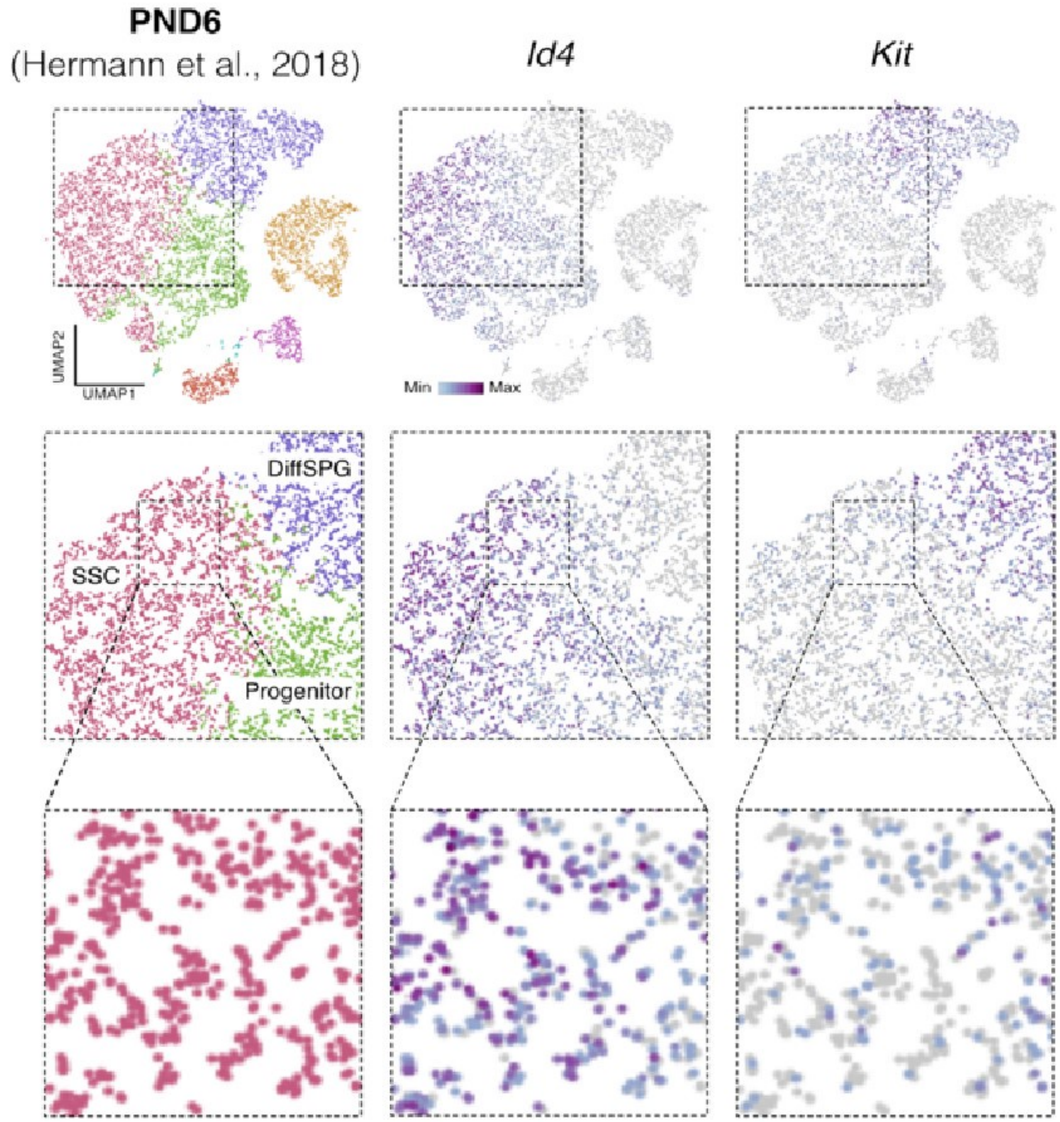
Author response image 4.

Deconvolution analysis of Sertoli bulk RNA-seq samples. Barplots of estimated cellular proportions for bulk RNAseq libraries reported in Thumfart et al., 2022. Expression matrices were derived from the analysis of single-cell RNA-seq datasets used to asses cellular composition of the SPGs bulk libraries.



Author response image 5.

Id4 and *Kit* are transcribed in SSCs. Seurat clusters from PND6 single-cell RNA-seq (left) and feature maps of gene expression for *Id4* (center) and *Kit* (right). Zoom in into SSCs (red).



Finally, regarding the following observation by the reviewer: "On the other hand, they report detection in the sorted population of markers such as c-KIT which is a well-known marker of differentiating spermatogonia, and that is in the same population in which ID4, a well-known marker of spermatogonial stem cells, was detected." It was recently shown using single-cell RNA that “nearly all differentiating spermatogonia at P3 (delineated as c-KIT+) are ID4-eGFP” (Law et al., 2019). While this finding does not exclude the fact that we have a mixture of SPGs cells, this finding supports the possibility that SPG cells express both markers of undifferentiated and differentiated cells, particularly in the early stages of postnatal development. Indeed, we observe that some cells labeled as SSC show signals for both *Id4* and *Kit* in single-cell RNA-seq data from Hermann et al., 2018 (Author response image 5).

Therefore, the results from the deconvolution analysis and our immunofluorescence data showing 85-95% PLZF+ cells in our cellular preparations underscore that our bulk RNA-seq libraries are mainly composed of SPGs. The deconvolution analysis also suggests a predominantly cellular composition of SSCs and to a lesser degree of differentiating SPGs. Our adult RNA-seq libraries show a small proportion of somatic cells (<0.10).

In the revised manuscript, we compiled the deconvolution analyses and present them in a condensed version in Supplementary Fig 2.

> *In general, the authors present observational data of the sort that is generated by RNA-seq and ATAC-seq analyses, and they speculate on the potential significance of several of these observations. However, they provide no definitive data to support any of their speculations. This further illustrates the fact that this study contributes little if any new information beyond that already available from the numerous previously published RNA-seq and ATAC-seq studies of spermatogenesis. In short, the study described in this manuscript does not advance the field.*

We acknowledge that RNA-seq and ATAC-seq datasets like ours are observational and that their interpretation can be speculative. Nevertheless, our datasets represent an additional useful resource for the community because they are comprehensive and high resolution, and can be exploited for instance, for studies in environmental epigenetics and epigenetic inheritance examining the immediate and long-term effects of postnatal exposure and their dynamics. The depth of our RNA sequencing allowed detect transcripts with a high dynamic range, which has been limited with classical RNA sequencing analyses of spermatogonial cells and with single-cell analyses (which have comparatively low coverage). Further, our experimental pipeline is affordable (more than single cell sequencing approaches) and in the case of adults, provides data per animal informing on the intrinsic variability in transcriptional and chromatin regulation across males. These points will be discussed in the revised manuscript.

> *In general, the authors present observational data of the sort that is generated by RNA-seq and ATAC-seq analyses, and they speculate on the potential significance of several of these observations. However, they provide no definitive data to support any of their speculations. This further illustrates the fact that this study contributes little if any new information beyond that already available from the numerous previously published RNA-seq and ATAC-seq studies of spermatogenesis. In short, the study described in this manuscript does not advance the field.*

Relevant information for both points was included in the Discussion of the revised manuscript.

> *The phenomenon of epigenetic priming is discussed, but then it seems that there is some expression of surprise that the data demonstrate what this reviewer would argue are examples of that phenomenon. The authors discuss the "modest correspondence between transcription and chromatin accessibility in SCs." Chromatin accessibility is an example of an epigenetic parameter associated with the primed state. The primed state is not fully equivalent to the actively expressing state. It appears that certain histone modifications along with transcription factors are critical to the transition between the primed and actively expressing states (in either direction). The cell types that were investigated in this study are closely related spermatogenic, and predominantly spermatogonial cell types. It is very likely that the differentially expressed loci will be primed in both the early (PND 8 or 15) and adult stages, even though those genes are differentially expressed at those stages. Thus, it is not surprising that there is not a strict concordance between +/- chromatin accessibility and +/- active or elevated expression.*

Relevant information was included in the Discussion of the revised manuscript.

> ***Reviewer #2:***
>
> *The objective of this study from Lazar-Contes et al. is to examine chromatin accessibility changes in "spermatogonial cells" (SCs) across testis development. Exactly what SCs are, however, remains a mystery. The authors mention in the abstract that SCs are*

> *undifferentiated male germ cells and have self-renewal and differentiation activity, which would be true for Spermatogonial STEM Cells (SSCs), a very small subset of total spermatogonia, but then the methods they use to retrieve such cells using antibodies that enrich for undifferentiated spermatogonia encompass both undifferentiated and differentiating spermatogonia. Data in Fig. 1B prove that most (85-95%) are PLZF+, but PLZF is known to be expressed both by undifferentiated and differentiating (KIT+) spermatogonia (Niedenberger et al., 2015; PMID: 25737569). Thus, the bulk RNA-seq and ATAC-seq data arising from these cells constitute the aggregate results comprising the phenotype of a highly heterogeneous mixture of spermatogonia (plus contaminating somatic cells), NOT SSCs. Indeed, Fig. 1C demonstrates this by showing the detection of Kit mRNA (a well-known marker of differentiating spermatogonia - which the authors claim on line 89 is a marker of SCs!), along with the detection of markers of various somatic cell populations (albeit at lower levels).*

The reviewer is correct that our spermatogonial cell populations are mixed and include undifferentiated and differentiated cells, hence the name of spermatogonia (SCs), and probably also contains some somatic cells. We acknowledge that this is a limitation of our isolation approach. To circumvent this limitation, we will conduct in silico deconvolution analysis using publicly available single-cell RNA sequencing datasets to obtain information about markers corresponding to undifferentiated and differentiated spermatogonia cells, and somatic cells. These additional analyses will provide information about the cellular composition of the samples and clarify the representation of undifferentiated and differentiated spermatogonial cells and other cells.

> *This admixture problem influences the results - the authors show ATAC-seq accessibility traces for several genes in Fig. 2E (exhibiting differences between P15 and Adult), including Ihh, which is not expressed by spermatogenic cells, and Col6a1, which is expressed by peritubular myoid cells. Thus, the methods in this paper are fundamentally flawed, which precludes drawing any firm conclusions from the data about changes in chromatin accessibility among spermatogonia (SCs?) across postnatal testis development.*

The reviewer raises concern about the lack of correspondence between chromatin accessibility and expression observed for some genes, arguing that this precludes drawing firm conclusions. However, a dissociation between chromatin accessibility and gene expression is normal and expected since chromatin accessibility is only a readout of protein deposition and occupancy e.g. by transcription factors, chromatin regulators, or nucleosomes, at specific genomic loci that does not give functional information of whether there is ongoing transcriptional activity or not. A gene that is repressed or poised for expression can still show a clear signal of chromatin accessibility at regulatory elements. The dissociation between chromatin accessibility and transcription has been reported in many different cells and conditions (PMID: 36069349, PMID: 33098772) including in spermatogonial cells (PMID: 28985528) and in gonads in different species (PMID: 36323261). Therefore, the dissociation between accessibility and transcription is not a reason to conclude that our data are flawed.

> *In addition, there already are numerous scRNA-seq datasets from mouse spermatogenic cells at the same developmental stages in question.*

This is true but full transcriptomic profiling like ours on cell populations provides different transcriptional information that is deeper and more comprehensive. Our datasets identified >17,000 genes while scRNA-seq typically identifies a few thousand of genes. Our analyses also identified full-length transcripts, variants, isoforms, and low abundance transcripts. These datasets are therefore a valuable addition to existing scRNAseq.

> *Moreover, several groups have used bulk ATAC-seq to profile enriched populations of spermatogonia, including from synchronized spermatogenesis which reflects a high degree of purity (see Maezawa et al., 2018 PMID: 29126117 and Schlief et al., 2023 PMID: 36983846 and in cultured spermatogonia - Suen et al., 2022 PMID: 36509798) - so this topic has already begun to be examined. None of these papers was cited, so it appears the authors were unaware of this work.*

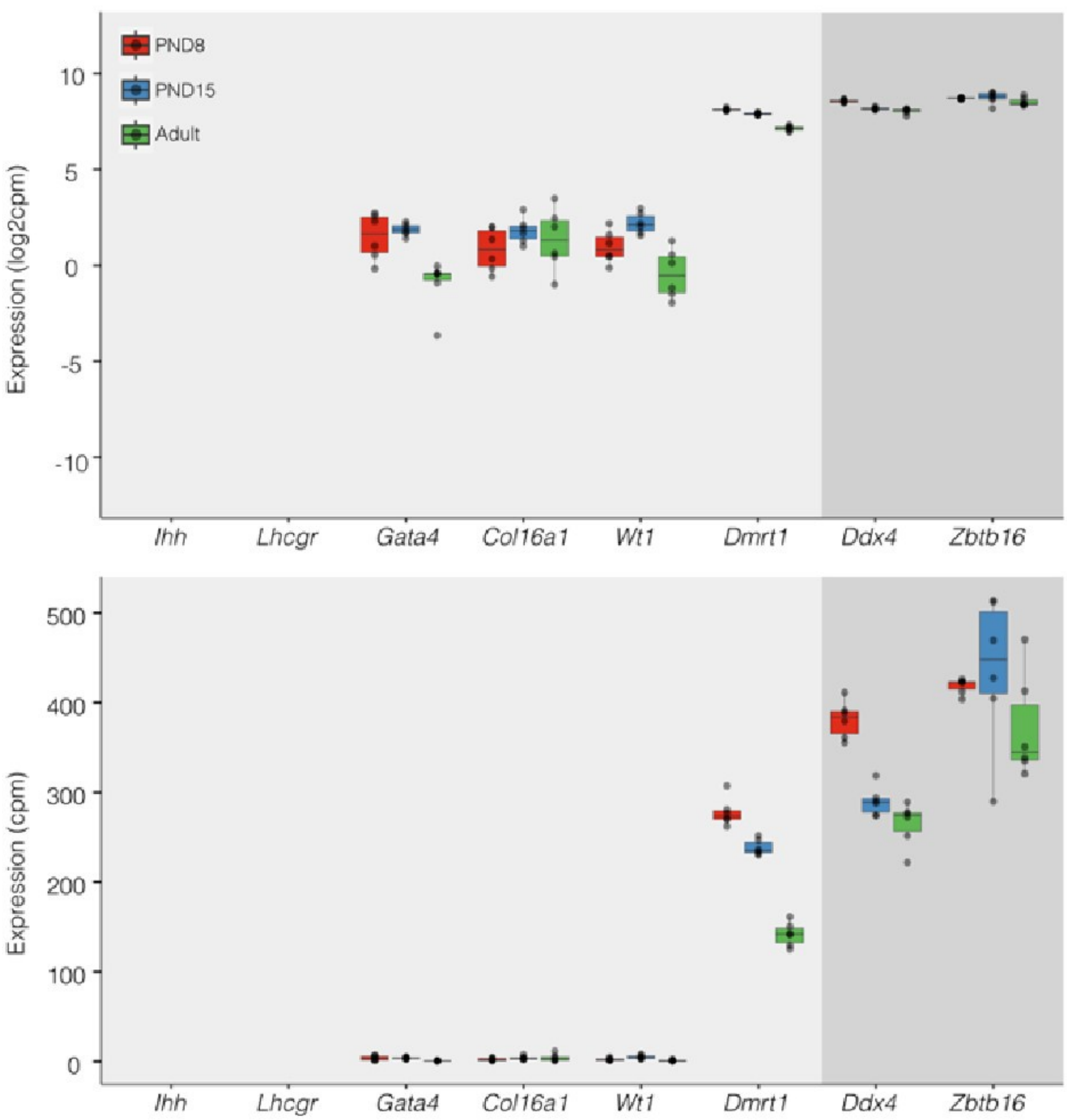
We apologize for not mentioning these studies in our manuscript, we will do so in the revised version.

> *The authors' methodological choice is even more surprising given the wealth of single-cell evidence in the literature since 2018 demonstrating the exceptional heterogeneity among spermatogonia at these developmental stages (the authors DID cite some of these papers, so they are aware). Indeed, it is currently possible to perform concurrent scATAC-seq and scRNA-seq (10x Genomics Multiome), which would have made these data quite useful and robust. As it stands, given the lack of novelty and critical methodological flaws, readers should be cautioned that there is little new information to be learned about spermatogenesis from this study, and in fact, the data in Figures 2-5 may lead readers astray because they do not reflect the biology of any one type of male germ cell. Indeed, not only do these data not add to our understanding of spermatogonial development, but they are damaging to the field if their source and identity are properly understood. Here are some specific examples of the problems with these data:*
>
> *Fig. 2D - Gata4 and Lhcgr are not expressed by germ cells in the testis.*
>
> *Fig. 3A - WT1 is expressed by Sertoli cells, so the change in accessibility of regions containing a WT1 motif suggests differential contamination with Sertoli cells. Since Wt1 mRNA was differentially high in P15 (Fig. 3B) - this seems to be the most likely explanation for the results. How was this excluded?*
>
> *Fig. 3D - Since Dmrt1 is expressed by Sertoli cells, the "downregulation" likely represents a reduction in Sertoli cell contamination in the adult, like the point above. Did the authors consider this?*

Regarding concerns about contamination by somatic cells (Transcription). In addition to the results of our deconvolution analysis (see response to Reviewer #1), we addressed the specific concern of the *paradoxical expression* of genes considered markers of somatic cells in the testis. For instance, we plotted the expression values of *Ihh, Lhcgr, Gata4, Col16a, Wt1,* and *Dmrt1* along with the expression values of *Ddx4* and *Zbtb16*. We observe that the expression level of *Ddx4* and *Zbtb16*, genes expressed predominantly in SPGs, is orders of magnitude higher than the one observed for the rest of the genes with the notable exception of *Dmrt1* which is also highly expressed (Fig.6). Indeed, our analysis of publicly available single-cell RNA-seq datasets shows that *Dmrt1* is robustly expressed in germ cells (Author response image 7), and as also noted by the reviewer, in Sertoli cells in postnatal stages. Notably, we observe a significant stepwise decrease in the expression of *Dmrt1* across the postnatal maturation of SPG cells. This is highly unlikely to be a result of major contamination by Sertoli cells of just our postnatal libraries. We based this statement on three observations. First, the deconvolution analysis of all our RNA-seq libraries using four different expression signature matrices from high-quality single-cell RNAseq from testis showed that our libraries are largely derived from SPGs. Second, the evaluation of our adult libraries with the PND6 signature matrix from Green et al., 2018 suggested that the proportion of Sertoli cells in our adult libraries, if any, would be higher than in our postnatal libraries (Author response image 3d, blue bars). This makes it unlikely that the observed decrease in expression of *Dmrt1* in adult samples is due to prominent somatic contamination of the postnatal libraries. Third, the step-wise decrease in *Dmrt1* expression seems to correlate with progression during postnatal development (Author response image 7) as feature maps of *Dmrt1* expression derived from public single-cell RNA-seq experiments show a reduction in expression in adult SPGs in comparison with early postnatal stages (Author response image 7 last two panels). Then, the observed effects are likely the result of developmental gene regulatory processes that operate during the developmental maturation of SPGs.

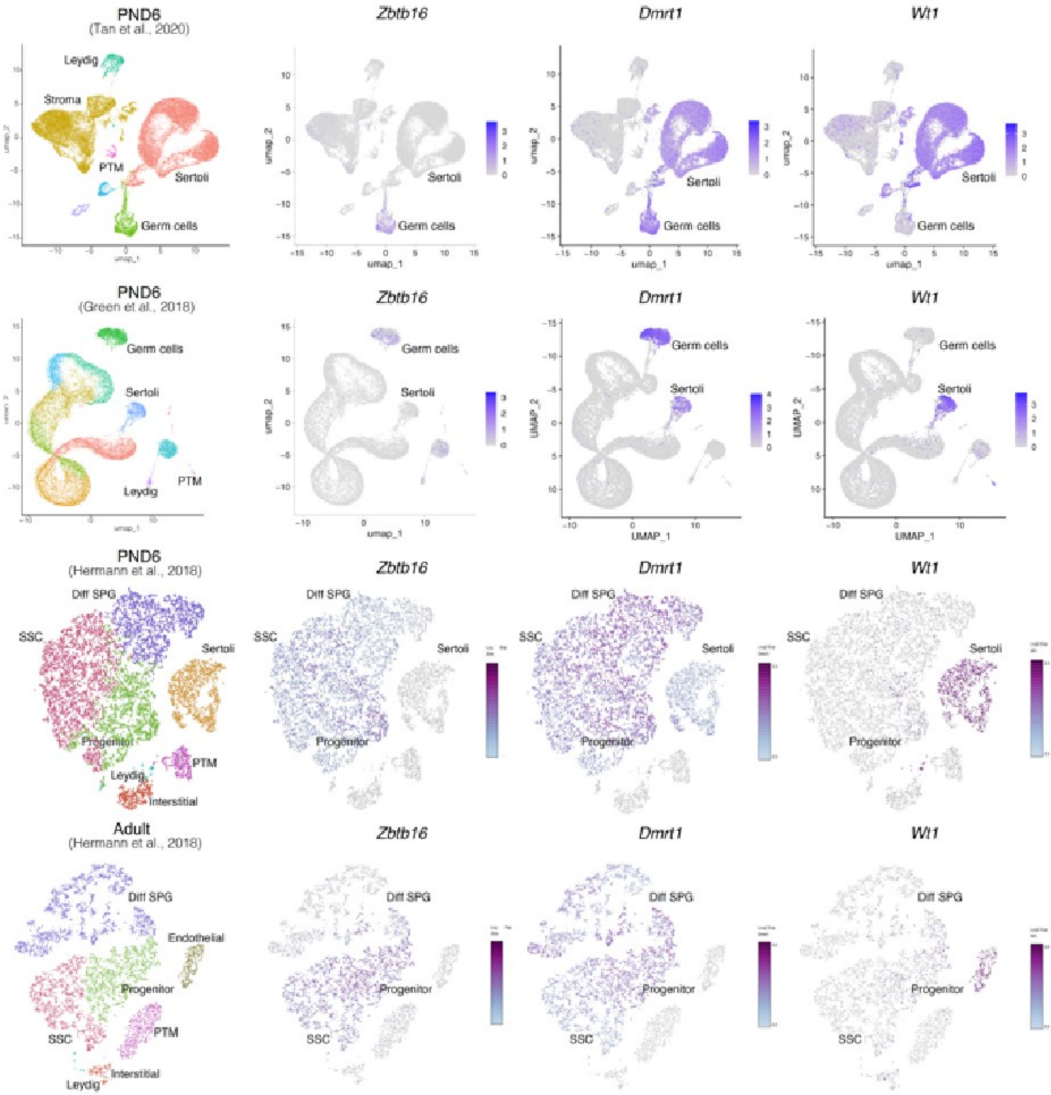
Author response image 6.

Expression of germ and somatic cell markers in our RNA-seq datasets. Boxplots of log2(CPM) (Top) and CPM (Bottom) values for selected genes from our RNAseq datasets. Each point in boxplots represent the expression value of a biological replicate.



Author response image 7.

Expression of germ and somatic cell markers in publicly available single-cell RNA-seq datasets. Seurat clusters from all analyzed single-cell RNA-seq datasets (first column from left) and feature maps of gene expression for *Zbtb16*, *Dmrt1* and *Wt1*.

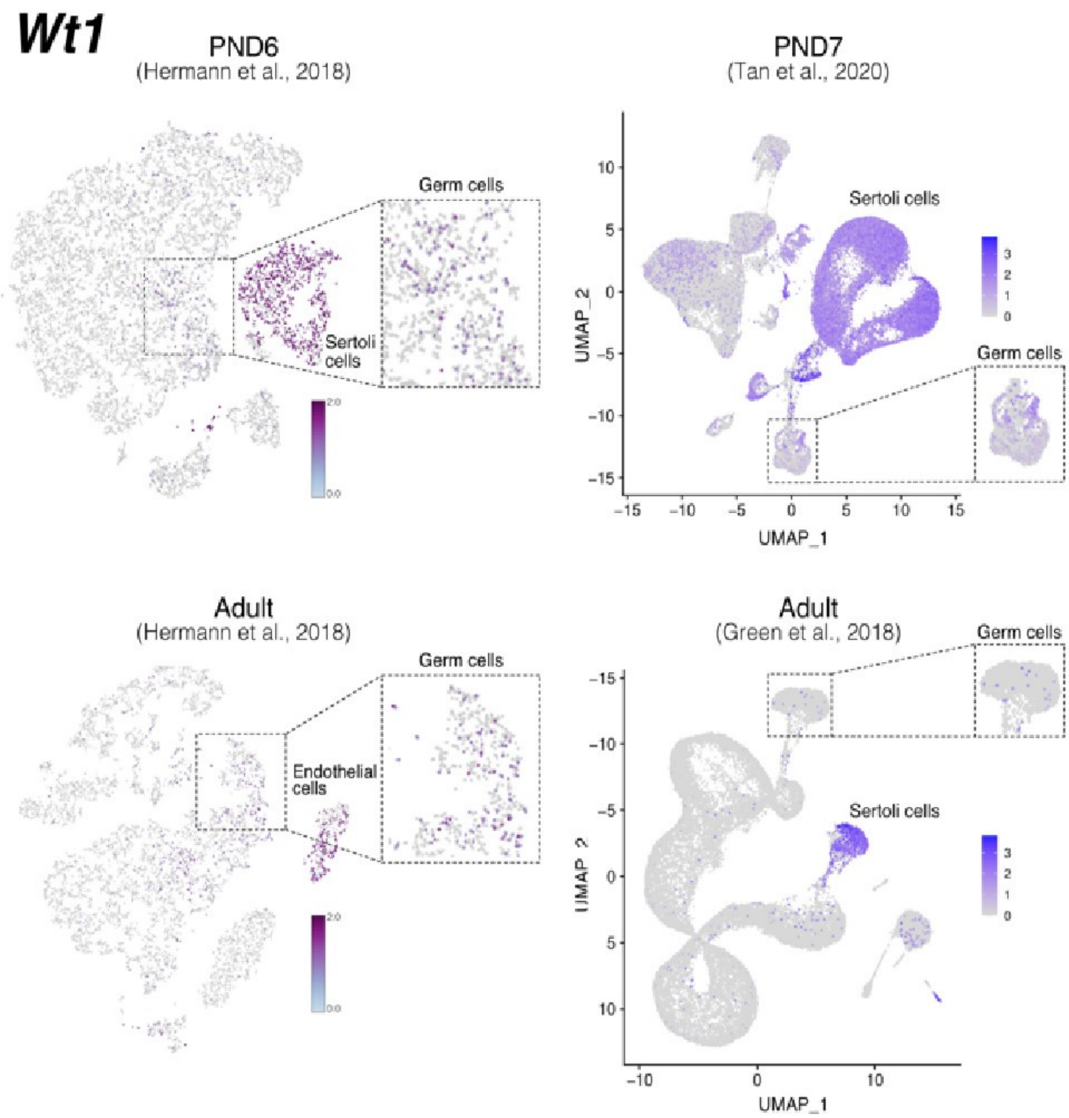


Consistent with the reviewer's observation, *Ihh* is not expressed in germ cells and indeed we do not detect signal at this locus nor *Lhcgr*. Furthermore, while we indeed observe a significant increase in the expression of *Wt1* in PND15 samples, its expression level is considerably lower than that of SPG markers. This is even more evident when plotting expression data in a linear scale rather than as a log2 transformation of the expression values. Whether such transcriptional profiles reflect developmentally regulated transcription, stochastic effects on gene expression, or potential somatic contamination is difficult to determine. However, based on our deconvolution data we believe it is unlikely that major contamination could account for our observations.

Notably, while *Wt1* is robustly expressed in nearly all Sertoli cells across postnatal development (Author response image 7), it is also detected in other cell types including SPGs - although in fewer cells and with lower expression levels-, consistent with our observations (Author response image 6 and 8). Therefore, the assignment of a gene as a marker of a particular cell type does not imply that such a gene is expressed *uniquely* in such cell, rather it is expressed in more cells and likely at higher levels.

Author response image 8.

Expression of *Wt1* in publicly available single-cell RNA-seq datasets. Feature maps of gene expression for *Wt1*. In dashed boxes, a zoom-in into germ cells cluster that show expression of *Wt1* at some of these cells.



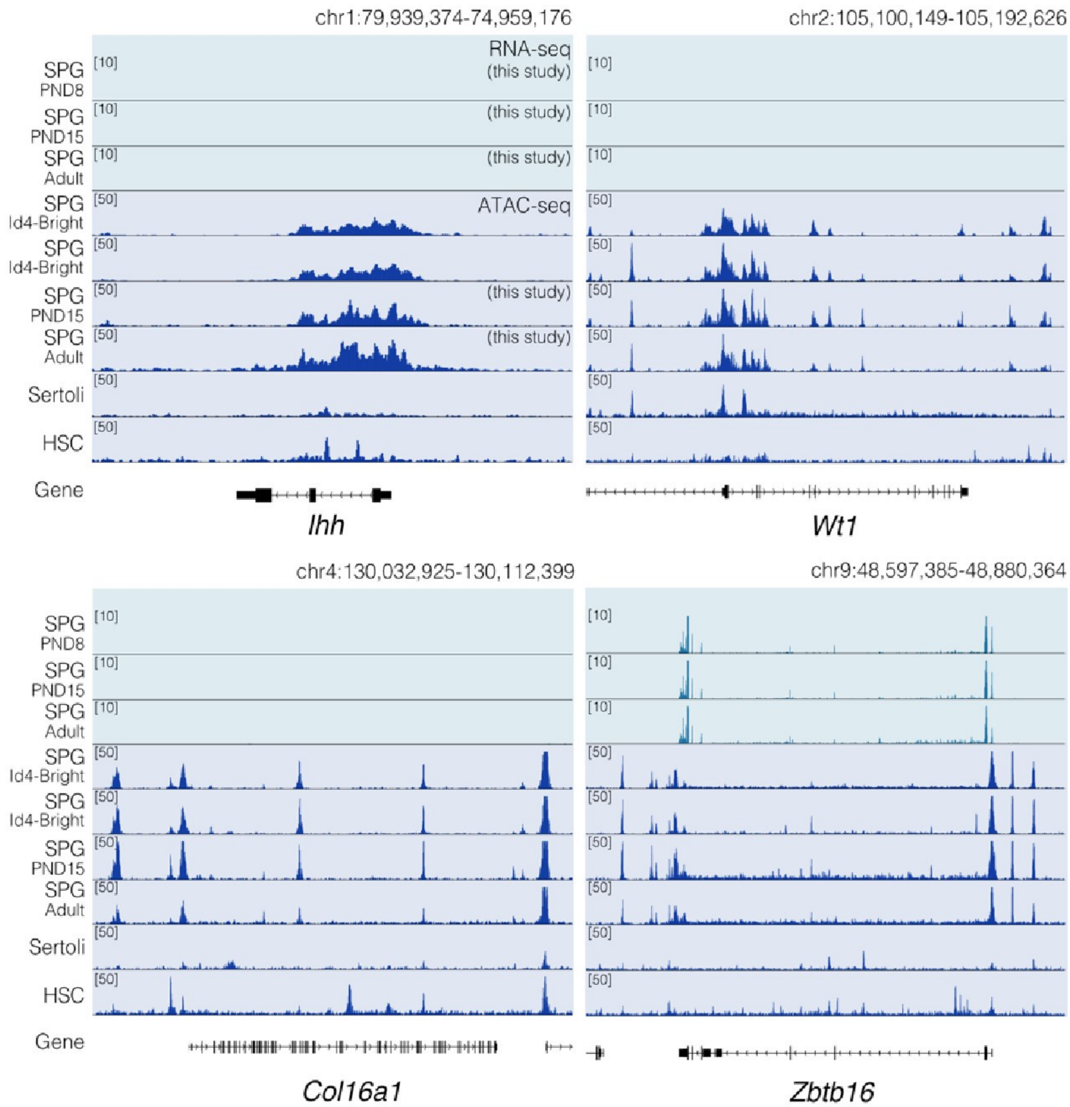
Regarding concerns about contamination by somatic cells (chromatin accessibility). In Figure 2 of our manuscript, we show the chromatin accessibility landscape of different genes, including genes either not expressed in testicular cells (*Ihh*) and those believed to be expressed exclusively in somatic cells (*Lhcgr, Gata4, Col16a1, Wt1*). For some of these genes, we reported changes in chromatin accessibility at specific sites between PND15 and adults (e.g. *Wt1* and *Col16a1*). The observation of "traces of chromatin accessibility" at these loci and the reported changes in accessibility raised concerns of potential contamination which "fundamentally flaw" our results, as stated by the reviewer. While we acknowledge that all enrichment methods have a margin of potential contamination, we fundamentally disagree with the reviewer's observations.

The term chromatin accessibility can be misleading. In principle, the term accessibility might suggest the literal lack of protein deposition at a given place in the genome. Rather, chromatin accessibility as evaluated by ATAC- seq (as in this case) must be interpreted as a measure of protein occupancy genome-wide (PMID: 30675018). Depending on the type of fragments analyzed we can obtain information regarding the occupancy of transcription factors (TFs), nucleosomes, and other chromatin-associated proteins that are present at genomic locations at a given time within a population of cells. The detection of chromatin accessibility at a given locus does not necessarily indicate transcription of the gene in a given cell type. A gene can be repressed or poised for expression and still show a clear signal of chromatin accessibility at its regulatory elements or along the gene body. For instance, in agreement with the reviewer's observation, neither *Ihh* nor *Lhcgr* is expressed in our datasets (Author response image 6 and Author response image 9), however, they show a distinctive pattern of chromatin accessibility in our datasets and publicly available ATAC-seq data derived from undifferentiated (Id4bright) and differentiating SPGs (Id4-dim) (Cheng et al., 2020) (Author response image 9). A similar argument can be applied regarding other loci such as *Wt1* and *Col6a1* for which we also observe extremely low levels of transcription. Therefore, the lack of transcription does not exclude that these loci display clear patterns of chromatin accessibility (Author response image 9). Notably, while traces of chromatin accessibility can also be observed in ATAC-seq datasets from embryonic Sertoli cells (Garcia-Moreno et al., 2019) and other somatic stem cells (hematopoietic stem cells; HSCs) (Xiang et al., 2020) (Author response image 9), the pattern of chromatin accessibility markedly differs with that observed in SPG cells. Therefore, the observed changes in chromatin accessibility are unlikely to result from contaminating somatic cells.

To strengthen our observation, we identified regions of chromatin accessibility in SPGs, Sertoli, and HSCs using both our datasets and publicly available ATAC-seq datasets. Overlap analysis revealed at least four groups of ATAC-seq peaks: 1) peaks shared among all analyzed cell types, 2)peaks shared just among SPG cells, 3) peaks specific to Sertoli cells and 4) peaks specific to HSCs (Author response image 10). Peaks shared among all tested cell-types are predominantly located at promoters of genes involved in translation and DNA replication (GO analysis adj p-value<0.05). In contrast, cell-type specific peaks are localized at intergenic and intragenic regions, suggesting localization at enhancer elements (Author response image 10). Indeed, GO analysis of cell-type specific peaks revealed enrichment for genes involved in male meiosis for SPGs, vesicle-mediated transport for Sertoli cells and in immune system process for HSCs, consistent with cell-type specific functions. If contamination by somatic cells, such as Sertoli cells, would be prominent as stated by the reviewer, we would expect to observe prominent ATAC-seq signal from our datasets at peaks specific to Sertoli cells. Notably, we don't observe ATAC-seq signal at peaks specific for Sertoli cells using our ATAC-seq samples. However, we observe robust signals at shared peaks and peaks specific to SPG cells. This observation, strongly argues against the possibility of major contamination by somatic cells.

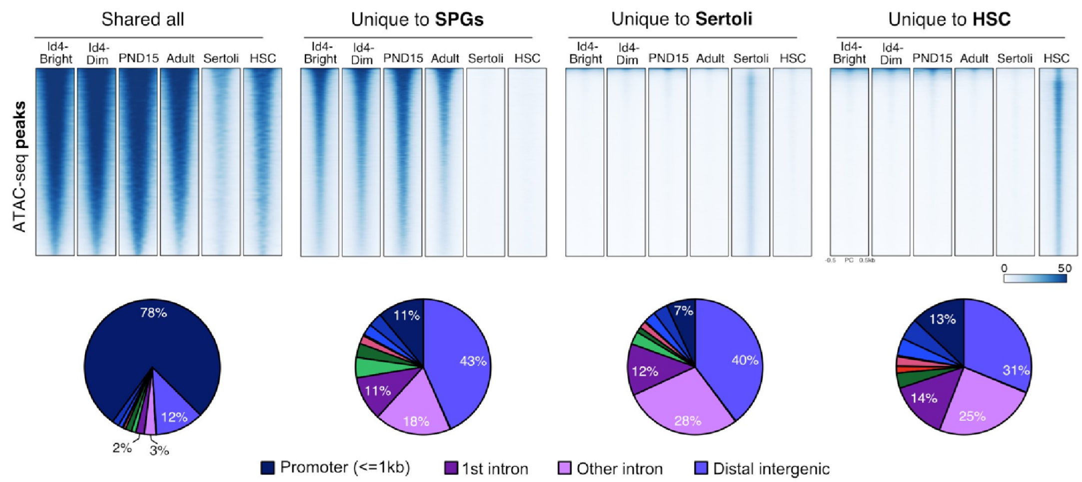
Author response image 9.

Chromatin accessibility profiles at specific loci differ between SPG cells and other cell types. Genome-browser tracks for *Ihh*, *Wt1*, *Col16a1* and *Zbtb16*. For each gene, an extended locus view is presented with RNA-seq data (this study) and normalized ATAC-seq tracks from our study and public sources (SPG Id4; GSE131657; Sertoli; GSM3346484; HSC; ENCFF204JEE). Public ATAC-seq datasets were generated enrichment methods similar to the one employed in our study.



Author response image 10.

Shared and cell-type specific ATAC-seq peaks among SPGs, Sertoli and HSC. *Up,* Normalized ATACseq signal heatmaps of shared and unique ATAC-seq peaks. PND15 and Adult samples are derived from our study. ATAC-seq signal is plotted +/- 500bp from peak center. *Bottom,* pie charts of ATAC-seq peaks genomic distribution.



> ***Reviewer #3:***
>
> *In this study, Lazar-Contes and colleagues aimed to determine whether chromatin accessibility changes in the spermatogonial population during different phases of postnatal mammalian testis development. Because actions of the spermatogonial population set the foundation for continual and robust spermatogenesis and the gene networks regulating their biology are undefined, the goal of the study has merit. To advance knowledge, the authors used mice as a model and isolated spermatogonia from three different postnatal developmental age points using a cell sorting methodology that was based on cell surface markers reported in previous studies and then performed bulk RNA-sequencing and ATAC-sequencing. Overall, the technical aspects of the sequencing analyses and computational/bioinformatics seem sound but there are several concerns with the cell population isolated from testes and lack of acknowledgment for previous studies that have also performed ATACsequencing on spermatogonia of mouse and human testes. The limitations, described below, call into question the validity of the interpretations and reduce the potential merit of the findings. I suggest changing the acronym for spermatogonial cells from SC to SPG for two reasons. First, SPG is the commonly used acronym in the field of mammalian spermatogenesis. Second, SC is commonly used for Sertoli Cells.*

We thank the reviewer for the suggestion and will rename SCs into SPG cells in the revised manuscript.

> *The authors should provide a rationale for why they used postnatal day 8 and 15 mice.*

We will provide a rationale for the use of postnatal 8 and 15 stages in the revised manuscript. Briefly, these stages are interesting to study because early to mid postnatal life is a critical window of development for germ cells during which environmental exposure can have strong and persistent effects. The possibility that changes in germ cells can happen during this period and persist until adulthood is an important area of research linked to disciplines like epigenetic toxicology and epigenetic inheritance.

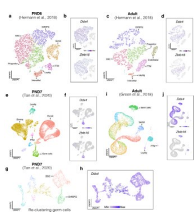
> *The FACS sorting approach used was based on cell surface proteins that are not germline-specific so there were undoubtedly somatic cells in the samples used for both RNA and ATAC sequencing. Thus, it is essential to demonstrate the level of both germ cell and undifferentiated spermatogonial enrichment in the isolated and profiled cell populations. To achieve this, the authors used PLZF as a biomarker of undifferentiated spermatogonia. Although PLZF is indeed expressed by undifferentiated spermatogonia,*

> *there have been several studies demonstrating that expression extends into differentiating spermatogonia. In addition, PLZF is not germ-cell specific and single-cell RNA-seq analyses of testicular tissue have revealed that there are somatic cell populations that express Plzf, at least at the mRNA level. For these reasons, I suggest that the authors assess the isolated cell populations using a germ-cell specific biomarker such as DDX4 in combination with PLZF to get a more accurate assessment of the undifferentiated spermatogonial composition. This assessment is essential for the interpretation of the RNA-seq and ATAC-seq data that was generated.*

In agreement with the reviewer's observation, *Zbtb16* (PLZF) is expressed in germ cells but also in somatic cells, in particular in the dataset derived from Green et al., 2018 (Author response image 11). However, when evaluating the expression patterns of *Ddx4*, we noticed that similar to *Zbtb16*, it is expressed both in the germ line and in the somatic compartment (Author response image 11). Notably, we observe expression of *Ddx4* in SSC but also in progenitors and differentiating SPGs (Author response image 11g). These observations suggest that at least at the transcript level, both genes are transcribed in germ cells and to a lesser degree in somatic cells.

Author response image 11.

Single-cell expression of *Ddx4* and *Zbtb16*. Seurat clusters from all analyzed single-cell RNA-seq datasets (a,c,e,g,i) and feature maps of gene expression for *Ddx4* and *Zbtb16* (b,d,f,j, h).



Finally, our deconvolution analysis using geneexpression signature matrices for different cellular populations suggest that our RNA-seq and ATAC-seq libraries are largely derived from SPG cells and in particular of SSCs.

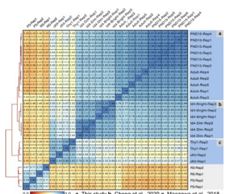
Furthermore, while this analysis suggested the presence of somatic cells, their proportion is minimal in comparison with germ cells (Author response images 1-4). This is also supported by ATAC-seq analysis of somatic cells from testis (Author response images 9 and 10).

> *A previous study by the Namekawa lab (PMID: 29126117) performed ATAC-seq on a similar cell population (THY1+ FACS sorted) that was isolated from pre-pubertal mouse testes. It was surprising to not see this study referenced in the current manuscript. In addition, it seems prudent to cross-reference the two ATAC-seq datasets for commonalities and differences. In addition, there are several published studies on scATACseq of human spermatogonia that might be of interest to cross-reference with the ATAC-seq data presented in the current study to provide an understanding of translational merit for the findings.*

We compared our ATAC-seq datasets with the ones from (Maezawa et al., 2017) and those from (Cheng et al., 2020). All these datasets were generated from FACSs sorted cells enriched for undifferentiating and differentiating SPGs. Sequencing files from Cheng et al, 2020 were equally processed as described in out methods section, while our pipeline was adjusted to process files from Maezawa et al., 2018 as they were single-end sequencing files. We generated a reference set of peaks from SPGs and calculated signal scores for all peaks across all samples. Then, calculated the Pearson correlation for all pairwise comparisons and generated a heatmap of correlations (Author response image 12). Two clusters emerge that separate the SPG samples from the pachytene spermatocytes and round spermatids reported by Maezawa et al., 2018. As expected SPG samples clustered together based on study of origin. Consistently, our postnatal samples formed one cluster next to but separated from the adult one. Similarly, the id4-bright samples clustered together and next to the id4-sim and the sample applied for the Thy1 and cKit samples. Notably, our samples and the ones from Cheng et al., 2020 have a higher correlation with each other when compared with the ones from Maezawa et al., 2018. Given the fundamental difference in library sequencing (single-end instead of the widely used paired-end for ATAC-seq experiments) we reasoned a comparison with the Maezawa et al., 2018 datasets is not optimal. Therefore, this data in addition to the one presented before (see response to Reviewer 1 and 2) strongly supports a predominantly SPG derivation of all our sequencing libraries.

Author response image 12.

Pearson correlation at the peak level among different ATAC-seq datasets. a) Our ATAC-seq libraries and ATAC-seq libraries from b) Cheng et al., 2020 and c) Maezawa et al., 2020. Thy1-1 and cKit libraries correspond to undifferentiated and differentiating SPGs, respectively. PS, pachytene spermatocytes and RS, round spermatids. Correlation analysis was done using Deeptools.



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